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Probiotics for the prevention of antibiotic-associated diarrhoea: A systematic review and meta-analysis

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PROBIOTICS FOR THE PREVENTION OF ANTIBIOTIC-ASSOCIATED DIARRHOEA: A SYSTEMATIC REVIEW AND META-ANALYSIS

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ABSTRACT

Objective: Antibiotics are one of the most commonly prescribed medications. They disturb normal gastrointestinal flora, which can lead to diarrhoea. Probiotics are non-pathogenic live micro-organisms thought to improve the microbial balance of the gastrointestinal tract and reduce colonisation by pathogenic bacteria. Our objective was to evaluate existing evidence for the use of probiotics in preventing antibiotic-associated diarrhoea in adults.

Design: Systematic review and meta-analysis.

Data sources: We performed a literature search of the electronic databases CINAHL Plus, EMBASE, MEDLINE (Ovid), and Web of Science from database inception to September 2019 as well as hand searching of trial registries and reference lists of related reviews.

Study selection: Two independent reviewers identified whether randomised controlled trials met the following inclusion criteria: adult population, a probiotic (any species, formulation or dose) intervention, placebo or no treatment control, incidence of antibiotic-associated diarrhoea as the outcome.

Data extraction and synthesis: Two reviewers independently collected data and assessed risk of bias using a pre-constructed data extraction form. For pooled data, if I^2 was less than 25% we used a fixed effects model; if I^2 was 25% or greater we used a random effects model. Sub-group analyses were performed to evaluate species-specific and dose-specific response.

Results: Thirty-seven studies (total 9799 participants) were included in this review. The pooled analysis suggests that co-administration of probiotics reduces the risk of antibiotic-associated diarrhoea in adults by 46% (Odds ratio (OR)=0.54 [95% CI: 0.44, 0.67], p<0.00001). In sub-group analyses, high dose compared with low dose of the same probiotic formulation demonstrated a positive protective effect (OR 0.53 [95% CI: 0.33, 0.84], p<0.01), and certain species particularly of the lactobacillus and bifidobacteria genera were found to be effective.

Conclusions: Our data suggests that probiotics are effective for preventing antibiotic-associated diarrhoea. Secondary analyses of higher dosages and certain species have shown increased effectiveness.

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STRENGTHS AND LIMITATIONS OF THIS STUDY

- Comprehensive literature search encapsulating seven years of new data since a comparable meta-analysis was last published by Cochrane
- Included only studies reporting on incidence of antibiotic-associated diarrhoea (AAD). Analysis of the effect of probiotics on duration of AAD may provide information about reducing severity.
- Significant heterogeneity between studies; significant variability in species and dose of probiotic interventions, start date and duration of co-administration of probiotics, baseline health of participants and antibiotics administered

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INTRODUCTION

Antibiotics are one of the most commonly prescribed medications worldwide. They disturb normal gastrointestinal flora [1] and a common consequence is diarrhoea [2]. This leads to prolonged hospital admissions [3], increased morbidity and mortality [4] and greater costs to the health system [5,6]. Antibiotic associated diarrhoea (AAD) has been reported in 5-35% of patients receiving antibiotics, varying with the type of antibiotics, the general health of the patient and their exposure to pathogens [7].

Probiotics are non-pathogenic live micro-organisms, thought to improve the microbial balance of the host and reduce colonisation by pathogenic bacteria. They can be delivered orally, adhere to target gastrointestinal epithelium and are stable in acid and bile (Dunne et al. 2001). Probiotics are increasingly available in capsules, powders and fermented milk drinks and are promoted in health food shops and supermarkets, and in the media, for a multitude of purposes.

It is suggested that probiotics are the panacea to AAD. This would be a very attractive solution to an expensive problem with significant morbidity. Several mechanisms are proposed to contribute to the protective effect of probiotics against AAD. Antibiotics are known to disrupt the complex balance of gastrointestinal flora, allowing colonisation by pathogenic organisms such as *Clostridium difficile* (*C. difficile*). Probiotics are believed to temporarily colonise the gut, producing bactericidal acids and peptides, and competing for nutrients and epithelial adhesion. Several species of the *Lactobacillus* genus are acid producing, lower environmental pH, and secrete exotoxins that inhibit pathogenic bacterial growth and inhibit binding of enterotoxins to gut epithelium [8–11]. Probiotics, particularly species in the *Lactobacillus* and *Saccharomyces* genera, are also reported to upregulate both the innate and adaptive immune systems through multiple mechanisms [12–15].

It is estimated that the increase in hospital admission due to AAD is 3.6 to 21.3 days [16,17]. One of the most expensive probiotics in Australia costs less than four dollars daily (VSL#3 AU$3.96 per day, available from NourishMeOrganics.com.au, April 2020). This is negligible compared to the cost of one day of Australian inpatient care, which according to 2017 data cost on average $2128 (Independent Hospital Pricing Authority, 2019).

Our hypothesis has been extensively investigated. In a Cochrane review [19] authors found moderate quality evidence for a large protective effect of probiotics against *Clostridium*

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difficile-associated diarrhoea (CDAD) (RR 0.36; 95% CI: 0.26, 0.51). Although *C. difficile* is the most common infectious organism isolated, in most cases of AAD no causative agent is found. Faecal analysis indicates incidence of *C. difficile* to be 20% [20], or less [21]. Furthermore while probiotics significantly reduce incidence of CDAD they do not appear to reduce *C. difficile* infection [19]. Goldenberg and colleagues suggest that the role of probiotics may be more in alleviating the symptoms of infection rather than preventing colonisation itself. For this reason, we examined the protective effect of probiotics against the occurrence of AAD without any specific causative organism.

The primary objective of this systematic review and meta-analysis was to evaluate existing evidence for the protective effect of probiotics against AAD in adults. Secondary objectives were to examine dose-specific and species-specific response; comparing different doses of the same probiotic and looking at the effectiveness different probiotic species in preventing AAD.

METHODS

Eligibility criteria

Randomised controlled trials (RCTs) comparing probiotic use (any strain, dose or formulation) to placebo, alternative dose, alternative probiotic strain, or no treatment, for the prevention of diarrhoea in adults receiving antibiotic therapy were eligible for inclusion in this review. Studies were excluded if they examined probiotics for treatment, rather than prevention, of AAD. Studies of adults, receiving any antibiotic therapy for a course of any length, or for any reason, were included. Studies of paediatric populations were excluded. The primary outcome measure was incidence of AAD, classified as three or more episodes of loose stools (5-7 Bristol Stool Scale), in any 24-hour period during the antibiotic treatment or follow-up phases.

Information sources and search strategy

A comprehensive electronic search (Appendix A) was performed of CINAHL Plus, EMBASE, MEDLINE (Ovid), and Web of Science from database inception to September 2019. The search was broad to include as many individual probiotic genera as possible and all clinical indications or outcomes (full search terms in Appendix A). Clinical trial registers including ClinicalTrials.gov and the Australia New Zealand Clinical Trials Registry (ANZCTR) were also searched, as well as hand searching of reference lists of relevant systematic reviews

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and included studies. The search was limited to RCTs in an adult population written in English and already in publication.

Study selection

Two of three of the authors (CG, GK and KL) independently assessed each study for inclusion in the review. Discrepancies were resolved through discussion.

Data collection process and data items

Data were extracted from each article by two of three of the authors (CG, GK and KL), using a standardised data extraction form following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines[22]. This was housed within Covidence online software (Covidence systematic review software, Veritas Health Innovation, Melbourne, Australia. Available at www.covidence.org). Data items collected included the number of patients allocated to each group, the probiotic species, formulation, dose and regime, whether a placebo or 'nothing' control was used, and outcome data on incidence of AAD.

Risk of bias in individual studies and risk of bias across studies

The quality of the evidence was assessed using the Cochrane Risk of Bias [23] tool within the Covidence online software. The domains assessed were allocation concealment, blinding of participants, personnel and outcome assessors, incomplete outcome data, selective outcome reporting and other sources of bias.

Summary measures and synthesis of results

The odds ratio (OR) was calculated for outcome measures for categorical data. A sensitivity analysis was performed, and the I^2 statistic was reported. Where the I^2 was less than 25% a fixed effects model was used, or if it was 25% or greater a random effects model was used.

Patient and public involvement

There was no patient or public involvement in this study.

RESULTS

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Study selection and characteristics

A total of 346 articles were identified in the initial search, of which 53 were eligible for full-text review, and 37 of those articles were eligible for inclusion in the study. Thirteen of these were published after the Cochrane review mentioned previously [24]. The PRISMA flow-diagram (figure 1) displays the study selection process and the citations and characteristics of each included study are presented in table 1.

Table 1: Characteristics of included studies

Reference	Total participants (N probiotic: N control)	Health care setting & age of patients	Probiotic species	Daily probiotic dose (cfu)	Duration of probiotic therapy	Control
[25]	2941 (1470:1471)	Inpatient ≥65	L. acidophilus, B. bifidum	60 x 10^9	21 days	Placebo
[26]	89 (44:45)	Inpatient 54-85	L. acidophilus, L. casei (Bio-K+ CL1285, Bio-K+ International Inc, Canada)	50 x 10^9	Duration of antibiotic	Placebo
[27]	202 (105:97)	Inpatient >18	L. acidophilus, L. bulgaricus, S. thermophilus	0.002 x 10^9	8 days	Nothing
[28]	151 (73:78)	Inpatient 25-50	S. boulardii	Not noted	Not noted	Placebo
[29]	343 (176:167)	Outpatient >18	L. acidophilus LA-5®, B. animalis ssp. lactis BB-12®	4 x 10^9	Duration of antibiotic	Placebo
[30]	23 (13:10)	Inpatient >18	L. reuteri ATCC 55730 (BioGaia Biologics, Sweden)	0.2 x 10^9	28 days	Placebo
[31]	124 (62:62)	Inpatient	S. boulardii	1g (cfu not noted)	14 days	Placebo
[32]	88 (30:29:29) (Probiotic: pasteurised: acidified)	Outpatients undergoing treatment for H. pylori	L. acidophilus, LA-5®, B. animalis, ssp. lactis, BB-12®and S. thermophilus	0.25 x 10^9		Pasteurised and acidified placebos
[33]	389 (204:185)	Outpatients undergoing treatment for H. pylori	S. boulardii	1g (cfu not noted)	14 days	Nothing
[34]	477 (246:231)	Inpatient 40-77	S. boulardii	36 x 10^9	Duration of antibiotic + 7 days	Placebo
[35]	160 (80:80)	Healthy 18-50	L. helveticus + L. rhamnosus	4 x 10^9	Duration of antibiotic + 7 days	Placebo
[36]	255 (86:85:84) (high: low: placebo)	Inpatient 50-70	L acidophilus CL1285® + L. casei LBC80R®	100 x 10^9, 50 x 10^9	Duration of antibiotic + 7 days	Placebo
[37]	79 (36:43)	Inpatient >18	L. acidophilus (gasseri), L. helveticus (bulgaricus) (Lactinex)	4 x 10^9	5 days	Placebo
[38]	85 (44:41)	Inpatient >18	L. casei Shirota (Yakult)	6.5 x 10^9	Duration of antibiotic + 7 days	Placebo
[39]	113 (57:56)	Inpatient >50	L. casei defensis, L. bulgaricus, S. thermophilus (Actimel)	0.21 x 10^9	Duration of antibiotic +5 days	Placebo

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[21]	271 (91:90:90) (Species1: species2: placebo)	Outpatient ≥45	(1) B. subtilis 3 & B. licheniformis 31, (2) B. licheniformis only	4 x 10^9	Duration of antibiotic + 7 days	Placebo
[40]	46 (26:20)	Inpatient	Lyophilized live L. acidophilus + B. bifidum (Infloran®)	6 x 10^9	14 days	Placebo
[41]	19 (5:7:7) (High: low: nothing)	Inpatient	C. Butyricum (CMB588)	0.02 x 10^9, 0.01 x 10^9	7 days	Nothing
[42]	40 (20:20)	Outpatients undergoing treatment for H. pylori	S. boulardii, Lactobacillus and B. longum	5 x 10^9	14 days	Nothing
[43]	38 (19:19)	Healthy 18-65	B. bifidum W23, B. lactis W18, B. longum W51, E. faecium W54, L. acidophilus W37 and W55, L. paracasei W72, L. plantarum W62, L. rhamnosus W71, L. salivarius W24	2 x 10^9	14 days	Placebo
[44]	69 (33:36)	Inpatient >18	S. boulardii	Not noted	Duration of antibiotic	Placebo
[45]	163 (80:83)	Inpatient >16	L. plantarum 299v	10 x 10^9	Duration of antibiotic + 7 days	Placebo
[46]	193 (97:96)	Inpatient >18	S. boulardii	Not noted	Duration of antibiotic + 3 days	Placebo
[47]	503 (168:168:167) (High: low: placebo)	Inpatient 30-70	L. acidophilus NCFM® (ATCC700396), L. paracasei Lpc-37 (ATCC SD5275), B. lactis Bl-07 (ATCC SD5220), B. lactis Bl-04 (ATCC SD5219)	17 x 10^9	Duration of antibiotic + 7 days	Placebo
[48]	678 (340:338)	Inpatient >18	L. casei Shirota (Yakult)	6.5 x 10^9	Duration of antibiotic + 3 days	Nothing
[49]	20 (11:9)	Inpatient >18	L. acidophilus, B. bifidum	20 x 10^9	20 days	Placebo
[50]	204 (106:98)	Inpatient >50	S. boulardii	10 x 10^9	Duration of antibiotic +7 days	Placebo
[51]	39 (23:16)	Inpatient >18	L. acidophilus	60 x 10^9	Duration of antibiotic + 14 days	Placebo
[52]	437 (216:221)	Inpatient >18	L. acidophilus + L. casei (Bio-K+ CL1285, Bio-K+ International Inc, Canada)	50 x 10^9	Duration of antibiotic + 5 days	Placebo
[53]	229 (117:112)	Inpatient >18	B. breve, B. longum, B. infants, L. acidophilus, L. plantarum, L. paracasei, L. delbrueckii subsp. bulgaricus, S. thermophilus (VSL#3)	900 x 10^9	Duration of antibiotic + 7 days	Placebo
[54]	214 (103:111)	Inpatient >18	L. rhamnosus R0011 + L. acidophilus R0052	4 x 10^9	14 days	Placebo
[55]	180 (90:90)	Inpatient >18	S. boulardii	Not noted	Duration of antibiotic + 14 days	Placebo
[56]	267 (133:134)	Inpatient >18	L. rhamnosus GG	20 x 10^9	14 days	Placebo
[57]	314 (122:125:67) (Probiotic: placebo: nothing)	Inpatient >18	L. acidophilus La-5, L. casei Lc-01 & B. lactis Bb-12	23 x 10^9	Duration of antibiotic + 5 days	Placebo and nothing control groups
[58]	63 (34:29)	Inpatient >18	L. rhamnosus GG (LGG), L. acidophilus La-5 and B. animalis subsp. lactis BB-12®	52.5 x 10^9	Not noted	Placebo

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[59]	158 (76:82)	Inpatient >18	L. casei Shirota (Yakult)	6.5 x 10^9	Duration of antibiotic + 7 days	Nothing
[60]	86 (40:46)	Inpatient >65	L. casei Shirota (Yakult)	13 x 10^9	Duration of admission	Placebo

All included studies were RCTs, conducted in adult populations. The age range was from 18 to 85 years. Interventions were comparable, in terms of all being probiotic formulations, however, there was variation in the dose and species of probiotic used as well as the probiotic formulation i.e. tablets, powder, yoghurt or fermented milk drink. We investigated these by sub-group analysis. All studies reported the outcome of interest; incidence of AAD as measured by three or more occurrences of diarrhoea in a 24-hour period, however, this was not the primary research question of the trial in all cases.

Risk of bias within studies

The risk of bias was assessed for all outcomes as described in the Cochrane Handbook [23]. The risk of bias summary is displayed in table 2. Overall 51% of studies were rated as having a low risk of bias, while 49% were at unclear risk of bias. No study was deemed to be at high risk of bias overall. Risk of bias due to missing data was considered to be very low given that all studies reported outcomes on incidence of AAD.

Table 2: Summary of risk of bias assessment

Risk of bias assessments	High quality Low risk of bias. No. (%)	Unclear quality Unclear risk of bias. No. (%)	Low quality High risk of bias. No. (%)
Sequence generation	20 (56)	15 (42)	2 (6)
Allocation concealment	16 (44)	21 (58)	0 (0)
Blinding of participants and personnel	21 (58)	13 (36)	3 (8)
Blinding of outcome assessors	20 (56)	17 (47)	0 (0)
Incomplete outcome data	23 (64)	14 (39)	0 (0)
Selective outcome reporting	21 (58)	15 (42)	1 (3)
Other sources of bias	24 (67)	13 (36)	0 (0)

a. *No. (%) is based on 37 randomised controlled trials that met inclusion criteria*

b. *Assessments made using the Cochrane Risk of Bias tool* [61]

Overall, 23 trials lacked adequate information to assess the overall risk of bias. Almost half of the RCTs reported only the probiotic genus and species that were used but not the strain (17/37), and many did not state whether treatment allocation was concealed (21/37) or failed

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to report an intention-to-treat analysis (21/37). However, 19 of the 37 trials reported that both participants and outcome assessors were blind to the intervention. In a further five studies either participants or assessors were blinded. There is some risk of bias due to clear lack of blinding in 8% of the trials and possibly in a further 41% of the trials which did not clearly report on blinding.

Results of individual studies and synthesis of results

Of the 37 studies included (table 1), 31 were placebo-controlled, seven provided 'no treatment' as the comparison - one study compared their treatment group with both a placebo group and a no treatment control [57]. Three trials compared two probiotic arms of differing dose with a control group [36,41,47]. Where there were two treatment groups (e.g. high dose and low dose) and one control group, the control group was split so that half served as comparator for each arm, in accordance with the Cochrane Systematic Review Guidelines [61]. Similarly in one study [32] where a probiotic group was compared with a placebo and 'no treatment' control, the probiotic group was split. Only one study was found to compare two strains of the same probiotic species, however, it was excluded as it was not randomised [62].

The overall pooled results of the 37 studies (reporting on 9799 participants) (Figure 2), comparing probiotics with a placebo or no treatment control, demonstrated a statistically significant reduction in the incidence of AAD by 46% (Odds ratio (OR)=0.54 [95% CI: 0.44, 0.67], p<0.00001). This suggests that the number needed to treat (NNT), to prevent one case of AAD would be 19 patients. There was significant heterogeneity (I^2=58%), therefore we used a random effects model for analysis.

Subgroup analyses

When sub-group analyses were performed, where probiotics were compared to placebo, and probiotics were compared to no treatment, both demonstrated a significant reduction in risk of diarrhoeal incidence of 39% (OR=0.61 [95% CI: 0.49, 0.75], p<0.00001) and 65% (OR 0.35 [95% CI: 0.21, 0.60], p<0.001) respectively. There was significant heterogeneity (I^2=50% and 59% respectively), therefore analyses used a random effects model.

Subgroup analysis: probiotic dose

Three studies directly compared high dose and low dose of the same probiotic [36,41,47]. The combined higher dose group demonstrated a significant reduction in the incidence of

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AAD by 47% (OR 0.53 [95% CI: 0.33, 0.84], p<0.01). All three studies included a third control group, so to include this data in the complete case analysis the control group was halved to create a comparator for each arm, as described above.

Subgroup analysis: probiotic species

Most studies used probiotic formulations containing more than one probiotic species so a sub-group analysis was performed on all individual species mentioned in included studies (Table 3). Where only one study existed, using a particular strain, we used that study's outcome data alone. The following probiotic strains demonstrated significant reduction in AAD when compared to control: *L. acidophilus, L. bulgaricus, L. casei, L. helveticus, L. paracasei, L. reuteri, L. rhamnosus, Lactobacillus spp., S. boulardii, B. longum, B. licheniformis,* and *B. subtilis.* However, *L. plantarum, B. animalis ssp. lactis, B. bifidum, S. thermophilus* and *C. butyricum* showed no improvement in AAD compared to control.

Table 3: Sub-group analyses by species or genus

Species	Studies (N)	Effect size
L. acidophilus	17 (5617)	38% (OR=0.62 [95% CI: 0.47, 0.82], p<0.001)
L. bulgaricus	4 (516)	68% (OR=0.32 [95% CI: 0.19, 0.55], p<0.0001)
L. casei	9 (1216)	49% (OR=0.51 [95% CI: 0.33, 0.79], p<0.01)
L. helveticus	2 (239)	45% (OR=0.55 [95% CI: 0.30, 1.01], p=0.05)
L. paracasei	3 (665)	57% (OR=0.43 [95% CI: 0.22, 0.86], p<0.05)
L. plantarum	3 (325)	Not significant
L. reuteri	1 (23)	92% (OR=0.08 [95% CI: 0.01, 0.91], p<0.05)
L. rhamnosus	5 (779)	45% (OR=0.55 [95% CI: 0.33, 0.91], p<0.05)
Lactobacillus spp.	27 (7722)	44% (OR=0.56 [95% CI: 0.43, 0.72], p<0.00001)
S. boulardii	9 (1827)	44% (OR=0.56 [95% CI: 0.37, 0.84], p<0.01)
B. animalis ssp. lactis	6 (1351)	Not significant
B. bifidum	4 (3165)	Not significant
B. longum	3 (202)	86% (OR=0.14 [95% CI: 0.05, 0.43], p<0.001)

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B. licheniformis	1 (136)	74% (OR=0.26 [95% CI: 0.09, 0.73], p<0.05)
B. subtilis	1 (271)	72% (OR=0.28 [95% CI: 0.14, 0.57], p<0.001)
S. thermophilus	4 (525)	Not significant
C. butyricum	1 (19)	Not significant

Adverse effects

No serious adverse events were reported in any of the included studies. Two studies reported a significantly greater number of non-serious adverse events in the probiotic group compared with the control group in their respective trials [29,57].

Small study effects

On inspection, the funnel plot of the complete case analysis (Figure 3) shows little evidence of publication bias. This suggests that our results are robust to theoretical large future trials with results in the opposite direction (i.e. that probiotics increase the risk of AAD rather than decreasing it).

DISCUSSION

Summary of evidence

The principal finding of this review is that probiotics significantly reduce the risk of AAD (OR=0.54 [95% CI: 0.44, 0.67], p<0.00001), with a NNT of 19 patients. These results were similarly positive in sub-group analyses, with high doses more effective than lower doses, and some species more effective than others. As is the nature of systematic reviews, we found significant heterogeneity between study designs and population, severity of illness of the study group, as well as probiotic species used, dose, timing and duration of probiotic administration.

Our results show evidence of a placebo effect, with a reduction in AAD incidence of 39% (OR=0.61 [95% CI: 0.49, 0.75], p<0.00001) when probiotics were compared to placebo, but 65% (OR 0.35 [95% CI: 0.21, 0.60], p<0.001) when probiotics were compared to no treatment. Placebo effect is not well understood but this could be attributed to expectations of symptom change with neurobiological underpinnings and actual effects on the brain and body [63]. This effect could be investigated further in well-designed, three-armed, RCTs.

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Secondary objectives included sub-group analyses of dose-response and species-specific response. In studies where dosages of the same species were compared [36,41,47], higher doses showed a significantly greater benefit (OR 0.53 [95% CI: 0.33, 0.84], p<0.01). It is difficult to interpret the dosage effect as the three studies that compared different doses of the same probiotic species made comparisons of differing magnitudes.

Many studies used blends of probiotic species, which made it difficult to establish independent associations. In our sub-group analyses, the individual probiotic species that demonstrated the greatest effect sizes were *B. longum* (3 studies, OR=0.14 [95% CI: 0.05, 0.43], p<0.001), *L. bulgaricus* (4 studies, OR=0.32 [95% CI: 0.19, 0.55], p<0.0001) and *L. paracasei* (3 studies, OR=0.43 [95% CI: 0.22, 0.86], p<0.05). Although not demonstrating the same magnitude of effect, *L. acidophilus* (17 studies, OR=0.62 [95% CI: 0.47, 0.82], p<0.001), *L. casei* (9 studies, OR=0.51 [95% CI: 0.33, 0.79], p<0.01) and *S. boulardii* (9 studies, OR=0.56 [95% CI: 0.37, 0.84], p<0.01) were used in many more studies. Overall, we have seen that the lactobacillus genus shows a good effect with a small confidence interval. The individual species *L. acidophilus* was used in the greatest number of studies, showing a significant effect with a relatively small confidence interval. A significant effect size was noted in some of the other species which may be worth looking into more specifically in future trials. *L. plantarum*, *B. animalis ssp. lactis*, *B. bifidum*, *S. thermophilus* and *C. butyricum* did not show evidence of effectiveness in reducing incidence of AAD.

One study [62] compared two strains of *L. casei,* Actimel® (*L. casei DN-114001*) and Yakult® (*L. casei Shirota*), for the prevention of AAD. In this study *L. casei DN-114001* was significantly more effective than *L. casei Shirota*, however, we were unable to include this data in our review as it was not randomised. Future RCTs should focus on direct comparisons of single or combined probiotic species.

The most commonly identified indication for antibiotics was treatment of *H. pylori*. In the five RCTs investigating adjunct probiotic use with antibiotics for *H.pylori* treatment we found a reduction in diarrhoeal side-effects (OR=0.43, [95% CI: 0.27, 0.69], p<0.001).

A treatment schedule was reported in most studies (34/37) with duration from five to 56 days. The largest proportion (13/37) of studies reported probiotic administration for the duration of antibiotic therapy plus a further seven days. As duration of antibiotic use also differed between studies there was quite a range in duration of probiotic use. It would,

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however, be biologically plausible to suggest that probiotic use should be continued for a period after the cessation of antibiotics. This is another question worth investigating in future trials.

Limitations

A limitation of this study is that the only outcome measured was incidence of diarrhoea. This review's utility in clinical practice may have been improved by also analysing duration of AAD. Additionally, it may have been informative to correlate outcomes with the duration of probiotic course or the date of probiotic initiation in relation to antibiotics.

This study was somewhat limited in scope as we did not include grey literature, correspondence with authors about trials currently in progress or to obtain missing data, and only included studies published in English.

Agreements and disagreements with other studies

This study is concordant with findings from the aforementioned Cochrane systematic review [19]. Goldenberg found that probiotics led to a significant reduction in incidence of AAD from 21% to 13% of participants (RR 0.60; 95% CI 0.49 to 0.72). The present review included 13 additional trials (7567 additional participants) published after the Cochrane review. This increase in total population increases the power of our review to detect smaller clinically relevant differences between groups in incidence of AAD.

Our sub-group analyses found similarly to one previous meta-analysis [64] that *Lactobacillus* and *Saccharomyces* genera reduced incidence of AAD. A more recent meta-analysis [65] found, also concordantly, that most *Lactobacillus* and *Bifidobacteria* species and *S. boulardii* were effective. Previous meta-analyses have not reported on dose-response.

Contrary to the findings of our review, one meta-analysis [66] reported that probiotics may not be effective in reducing the incidence of AAD, however, only 13 studies were analysed, including adults over 65 only. This suggests a need for looking separately at the effectiveness of probiotics in populations of younger and older adults.

CONCLUSIONS

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This review provides evidence of a protective effect of probiotics against AAD (OR=0.53 [95% CI: 0.42, 0.67], p<0.001). In absolute terms, the NNT, to prevent one case of AAD would be 19 patients. In studies that compared dosage of the same probiotic higher doses were found to be significantly more effective. The probiotic species that demonstrated the greatest effect were *B. longum*, *L. bulgaricus* and *L. paracasei* but were each only studied in a handful of trials. The species *L. acidophilus*, *L. casei* and *S. boulardii*, although demonstrating relatively smaller effect sizes, were used in many more trials focusing on prevention of AAD. The species that did not demonstrate a significant effect in the prevention of AAD were *L. plantarum*, *B. animalis ssp. lactis*, *B. bifidum*, *S. thermophilus* and *C. butyricum*.

This review has demonstrated overall effectiveness of probiotics for the prevention of AAD, including a dose-response and some species-specific effectiveness of probiotics. However, further high-quality trials are needed, focusing on direct comparison of doses, comparison of individual species and combinations of species. Future studies should record not only incidence of diarrhoea, but also duration and associated length of stay in hospital so that an assessment of the cost-benefit of probiotic therapy can be made. Trials should ensure that adequate blinding and reporting of adverse events is included.

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AUTHOR CONTRIBUTIONS

Study concept and design: CG, CH, KL
Acquisition of data: CG, GK, KL
Analysis and interpretation of data: CG, GK, KL
Drafting of the manuscript: CG, KL
Critical revision of the manuscript for important intellectual content: CG, CH, KL
Statistical analysis: CG, KL
Study supervision: KL

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COMPETING INTERESTS

None to declare

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ETHICS APPROVAL

This systematic review and meta-analysis was based on published data. As researchers did not access any information that could lead to the identification of an individual patient, no concerning ethical issue was raised in this research. Therefore, obtaining ethical approval and consent of participants was waived.

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REFERENCES

1 Francino MP. Antibiotics and the human gut microbiome: Dysbioses and accumulation of resistances. Front. Microbiol. 2016;6:1543. doi:10.3389/fmicb.2015.01543

2 Young VB, Schmidt TM. Antibiotic-associated diarrhea accompanied by large-scale alterations in the composition of the fecal microbiota. *J Clin Microbiol* 2004;42:1203–6. doi:10.1128/JCM.42.3.1203-1206.2004

3 Elseviers MM, Van Camp Y, Nayaert S, *et al.* Prevalence and management of antibiotic associated diarrhea in general hospitals. *BMC Infect Dis* 2015;15:129. doi:10.1186/s12879-015-0869-0

4 Bishara J, Peled N, Pitlik S, *et al.* Mortality of patients with antibiotic-associated diarrhoea: the impact of Clostridium difficile. *J Hosp Infect* 2008;68:308–14. doi:10.1016/J.JHIN.2008.01.033

5 Kamdeu Fansi AA, Guertin JR, LeLorier J. Savings from the use of a probiotic formula in the prophylaxis of antibiotic-associated diarrhea. *J Med Econ* 2012;15:53–60. doi:10.3111/13696998.2011.629015

6 Lenoir-Wijnkoop I, Nuijten MJC, Craig J, *et al.* Nutrition economic evaluation of a probiotic in the prevention of antibiotic-associated diarrhea. *Front Pharmacol* 2014;5 FEB. doi:10.3389/fphar.2014.00013

7 McFarland L V. Antibiotic-associated diarrhea: epidemiology, trends and treatment. *Future Microbiol* 2008;3:563–78. doi:10.2217/17460913.3.5.563

8 Naaber P, Mikelsaar M. Interactions between Lactobacilli and antibiotic-associated diarrhea. *Adv Appl Microbiol* 2004;54:231–60. doi:10.1016/S0065-2164(04)54009-1

9 Elmer GW, McFarland L V. Suppression by Saccharomyces boulardii of toxigenic Clostridium difficile overgrowth after vancomycin treatment in hamsters. *Antimicrob Agents Chemother* 1987;31:129–31. doi:10.1128/AAC.31.1.129

10 Pothoulakis C, Kelly CP, Joshi MA, *et al.* Saccharomyces boulardii inhibits Clostridium difficile toxin A binding and enterotoxicity in rat ileum. *Gastroenterology* 1993;104:1108–15.http://www.ncbi.nlm.nih.gov/pubmed/8462799 (accessed 7 Mar 2018).

11 Castagliuolo I, Riegler MF, Valenick L, *et al.* Saccharomyces boulardii protease inhibits the effects of Clostridium difficile toxins A and B in human colonic mucosa. *Infect Immun* 1999;67:302–7.http://www.ncbi.nlm.nih.gov/pubmed/9864230 (accessed 7 Mar 2018).

12 Link-Amster H, Rochat F, Saudan KY, *et al.* Modulation of a specific humoral immune response and changes in intestinal flora mediated through fermented milk intake. *FEMS Immunol Med Microbiol* 1994;10:55–63.http://www.ncbi.nlm.nih.gov/pubmed/7874079 (accessed 8 Mar 2018).

13 Qamar A, Aboudola S, Warny M, *et al.* Saccharomyces boulardii Stimulates Intestinal Immunoglobulin A Immune Response to Clostridium difficile Toxin A in Mice. *Infect Immun* 2001;69:2762–5. doi:10.1128/IAI.69.4.2762-2765.2001

14 Yasui H, Shida K, Matsuzaki T, *et al.* Immunomodulatory function of lactic acid bacteria.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Antonie Van Leeuwenhoek 1999;76:383–9.http://www.ncbi.nlm.nih.gov/pubmed/10532394 (accessed 8 Mar 2018).

15 Gill HS, Rutherfurd KJ, Cross ML. Dietary probiotic supplementation enhances natural killer cell activity in the elderly: an investigation of age-related immunological changes. *J Clin Immunol* 2001;21:264–71.http://www.ncbi.nlm.nih.gov/pubmed/11506196 (accessed 8 Mar 2018).

16 Wilcox MH, Cunniffe JG, Trundle C, *et al.* Financial burden of hospital-acquired Clostridium difficile infection. *J Hosp Infect* 1996;34:23–30.http://www.ncbi.nlm.nih.gov/pubmed/8880547 (accessed 8 Feb 2018).

17 Walk ST, Young VB. Emerging Insights into Antibiotic-Associated Diarrhea and Clostridium difficile Infection through the Lens of Microbial Ecology. *Interdiscip Perspect Infect Dis* 2008;2008:125081. doi:10.1155/2008/125081

18 Independent Hospital Pricing Authority. National Hospital Cost Data Collection: Cost Report: Round 19, Financial Year 2014-2015. 2016. https://www.ihpa.gov.au/sites/g/files/net636/f/publications/nhcdc-round18.pdf (accessed 29 Jan 2018).

19 Goldenberg J, Ma S, Saxton J, *et al.* Probiotics for the prevention of Clostridium difficile-associated diarrhea in adults and children. *Cochrane Libr* 2013;:CD006095. doi:10.1002/14651858.CD006095.pub3.www.cochranelibrary.com

20 Hurley BW, Nguyen CC. The Spectrum of Pseudomembranous Enterocolitis and Antibiotic-Associated Diarrhea. *Arch Intern Med* 2002;162:2177–84.https://pdfs.semanticscholar.org/cd35/9b31def5f11b39fd18f0657d24d146275ac9.pdf (accessed 1 Mar 2018).

21 Horosheva T V., Sorokulova I, Vodyanoy V. Efficacy of Bacillus probiotics in prevention of antibiotic-associated diarrhoea: a randomized, double-blind, placebo-controlled clinical trial. *JMM Case Reports* 2014;1:1–6. doi:10.1099/jmmcr.0.004036

22 Moher D, Liberati A, Tetzlaff J, *et al.* Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement (Reprinted from Annals of Internal Medicine). *PLoS Med* 2009;6:1–6. doi:10.1371/journal.pmed.1000097

23 Higgins J, Altman D. Chapter 8: Assessing risk of bias in included studies Table 8.5 a. In: *Cochrane Handbook for Systematic Reviews of …*. The Cochrane Collaboration, Availa 2008. 1–50. doi:10.1001/jama.2013.109501.Conflict

24 Goldenberg J, Ma S, Saxton J, *et al.* Probiotics for the prevention of Clostridium difficile-associated diarrhea in adults and children. *Cochrane Libr* 2013;:CD006095. doi:10.1002/14651858.CD006095.pub3.www.cochranelibrary.com

25 Allen SJ, Wareham K, Wang D, *et al.* Lactobacilli and bifidobacteria in the prevention of antibiotic-associated diarrhoea and Clostridium difficile diarrhoea in older inpatients (PLACIDE): a randomised, double-blind, placebo-controlled, multicentre trial. *Lancet* 2013;382:1249–57. doi:10.1016/S0140-6736(13)61218-0

26 Beausoleil M, Fortier N, Guénette S, *et al.* Effect of a fermented milk combining Lactobacillus

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

acidophilus CL1285 and Lactobacillus casei in the prevention of antibiotic-associated diarrhea: A randomized, double-blind, placebo-controlled trial. *Can J Gastroenterol* 2007;21:732–6.http://www.ncbi.nlm.nih.gov/pubmed/18026577 (accessed 6 Dec 2017).

27 Beniwal RS, Arena VC, Thomas L, *et al.* A randomized trial of yogurt for prevention of antibiotic-associated diarrhea. *Dig Dis Sci* 2003;48:2077–82. doi:10.1023/A:1026155328638

28 Can M, Beşirbellioglu BA, Avci IY, *et al.* Prophylactic Saccharomyces boulardii in the prevention of antibiotic-associated diarrhea: a prospective study. *Med Sci Monit* 2006;12:PI19-I22. doi:6635 [pii]

29 Chatterjee S, Kar P, Das T, *et al.* Randomised placebo-controlled double blind multicentric trial on efficacy and safety of Lactobacillus acidophilus LA-5 and Bifidobacterium BB-12 for prevention of antibiotic-associated diarrhoea. *J Assoc Physicians India* 2013;61:708–12.http://www.ncbi.nlm.nih.gov/pubmed/24772726 (accessed 6 Dec 2017).

30 Cimperman L, Bayless G, Best K, *et al.* A Randomized, Double-blind, Placebo-controlled Pilot Study of Lactobacillus reuteri ATCC 55730 for the Prevention of Antibiotic-associated Diarrhea in Hospitalized Adults. *J Clin Gastroenterol* 2011;45:785–9. doi:10.1097/MCG.0b013e3182166a42

31 Cindoruk M, Erkan G, Karakan T, *et al.* Efficacy and safety of Saccharomyces boulardii in the 14-day triple anti-Helicobacter pylori therapy: A prospective randomized placebo-controlled double-blind study. *Helicobacter* 2007;12:309–16. doi:10.1111/j.1523-5378.2007.00516.x

32 de Vrese M, Kristen H, Rautenberg P, *et al.* Probiotic lactobacilli and bifidobacteria in a fermented milk product with added fruit preparation reduce antibiotic associated diarrhea and Helicobacter pylori activity. *J Dairy Res* 2011;78:396–403. doi:10.1017/S002202991100063X

33 Duman DG, Bor S, Ozütemiz O, *et al.* Efficacy and safety of Saccharomyces boulardii in prevention of antibiotic-associated diarrhoea due to Helicobacterpylori eradication. *Eur J Gastroenterol Hepatol* 2005;17:1357–61.http://www.ncbi.nlm.nih.gov/pubmed/16292090 (accessed 6 Dec 2017).

34 Ehrhardt S, Guo N, Hinz R, *et al.* Saccharomyces boulardii to prevent antibiotic-associated diarrhea: A randomized, double-masked, placebo-controlled trial. *Open Forum Infect Dis* 2016;3:1–7. doi:10.1093/ofid/ofw011

35 Evans M, Salewski RP, Christman MC, *et al.* Effectiveness of Lactobacillus helveticus and Lactobacillus rhamnosus for the management of antibiotic-associated diarrhoea in healthy adults: a randomised, double-blind, placebo-controlled trial. *Br J Nutr* 2016;116:94–103. doi:10.1017/S0007114516001665

36 Gao XW, Mubasher M, Fang CY, *et al.* Dose-response efficacy of a proprietary probiotic formula of Lactobacillus acidophilus CL1285 and Lactobacillus casei LBC80R for antibiotic-associated diarrhea and Clostridium difficile-associated diarrhea prophylaxis in adult patients. *Am J Gastroenterol* 2010;105:1636–41. doi:10.1038/ajg.2010.11

37 Gotz V, Romankiewicz JA, Moss J, *et al.* Prophylaxis against ampicillin-associated diarrhea with a lactobacillus preparation. *Am J Hosp Pharm* 1979;36:754–7.http://ovidsp.ovid.com/ovidweb.cgi?T=JS&PAGE=reference&D=med1&NEWS=N&AN=1115

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

46

38 Helps A, Bell E, Mactier R. Prospective randomized double-blind study of efficacy of probiotic milk drink in reducing the incidence of antibiotic-associated diarrhoea and clostridium difficile Diarrhoea. *Int J Probiotics Prebiotics* 2015;11:145–52.

39 Hickson M, D'Souza AL, Muthu N, *et al.* Use of probiotic Lactobacillus preparation to prevent diarrhoea associated with antibiotics: randomised double blind placebo controlled trial. *BMJ* 2007;335:80. doi:10.1136/bmj.39231.599815.55

40 Iamharit J, Harnsomburana P. Prophylactic Use of Probiotic (Infloran ®) to Prevent Antibiotic Associated Diarrhea in Medical Ward Patients at Rajavithi Hospital. *Thai J Gastroenterol* 2010;11:155–60.

41 Imase K, Takahashi M, Tanaka A, *et al.* Efficacy of Clostridium butyricum preparation concomitantly with Helicobacter pylori eradication therapy in relation to changes in the intestinal microbiota. *Microbiol Immunol* 2008;52:156–61. doi:https://dx.doi.org/10.1111/j.1348-0421.2008.00026.x

42 Khanal A, Mishra A, Shrestha D, *et al.* Efficacy Of A Mixture Of Probiotic Containing saccharomyces boulardii and lactobacillus in Preventing Antibiotic Associated Diarrhea In Patients Taking Triple Therapy For H Pylori Eradication. *Post-Graduate Med J NAMS* 2015;15.https://pmjn.org.np/index.php/pmjn/article/view/268 (accessed 10 Jul 2019).

43 Koning CJM, Jonkers DMAE, Stobberingh EE, *et al.* The effect of a multispecies probiotic on the intestinal microbiota and bowel movements in healthy volunteers taking the antibiotic amoxycillin. *Am J Gastroenterol* 2008;103:178–89. doi:10.1111/j.1572-0241.2007.01547.x

44 Lewis SJ, Potts LF, Barry RE. The lack of therapeutic effect of Saccharomyces boulardii in the prevention of antibiotic-related diarrhoea in elderly patients. *J Infect* 1998;36:171–4. doi:10.1016/S0163-4453(98)80008-X

45 Lönnermark E, Friman V, Lappas G, *et al.* Intake of Lactobacillus plantarum Reduces Certain Gastrointestinal Symptoms During Treatment With Antibiotics. *J Clin Gastroenterol* 2010;44:106–12. doi:10.1097/MCG.0b013e3181b2683f

46 McFarland L V., Surawicz CM, Greenberg RN, *et al.* Prevention of β-Lactam-Associated Diarrhea by Saccharomyces boulardii Compared with Placebo. *Am J Gastroenterol* 1995;90:439–48. doi:10.1111/j.1572-0241.1995.tb08241.x

47 Ouwehand AC, DongLian C, Weijian X, *et al.* Probiotics reduce symptoms of antibiotic use in a hospital setting: a randomized dose response study. *Vaccine* 2014;32:458–63. doi:10.1016/j.vaccine.2013.11.053

48 Pirker A, Stockenhuber A, Remely M, *et al.* Effects of antibiotic therapy on the gastrointestinal microbiota and the influence of *Lactobacillus casei*. *Food Agric Immunol* 2013;24:315–30. doi:10.1080/09540105.2012.689816

49 Plummer S, Weaver MA, Harris JC, *et al.* Clostridium difficile pilot study: effects of probiotic supplementation on the incidence of C-difficile diarrhoea. *Int Microbiol* 2004;7:59–62.file:///Users/clarelouisagoodman/Downloads/09 Plummer.pdf

50 Pozzoni P, Riva A, Bellatorre AG, *et al.* Saccharomyces boulardii for the prevention of

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

antibiotic-associated diarrhea in adult hospitalized patients: a single-center, randomized, double-blind, placebo-controlled trial. *Am J Gastroenterol* 2012;107:922–31. doi:10.1038/ajg.2012.56

51 Safdar N, Barigala R, Said A, *et al.* Feasibility and tolerability of probiotics for prevention of antibiotic-associated diarrhoea in hospitalized US military veterans. *J Clin Pharm Ther* 2008;33:663–8. doi:10.1111/j.1365-2710.2008.00980.x

52 Sampalis J, Psaradellis E, Rampakakis E. Efficacy of BIO K+ CL1285 in the reduction of antibiotic-associated diarrhea - a placebo controlled double-blind randomized, multi-center study. *Arch Med Sci* 2010;6:56–64. doi:10.5114/aoms.2010.13508

53 Selinger CPP, Bell A, Cairns A, *et al.* Probiotic VSL#3 prevents antibiotic-associated diarrhoea in a double-blind, randomized, placebo-controlled clinical trial. *J Hosp Infect* 2013;84:159–65. doi:10.1016/j.jhin.2013.02.019

54 Song HJ, Kim J-Y, Jung S-A, *et al.* Effect of probiotic Lactobacillus (Lacidofil® cap) for the prevention of antibiotic-associated diarrhea: a prospective, randomized, double-blind, multicenter study. *J Korean Med Sci* 2010;25:1784–91. doi:10.3346/jkms.2010.25.12.1784

55 Surawicz CM, Elmer GW, Speelman P, *et al.* Prevention of antibiotic-associated diarrhea by Saccharomyces boulardii: a prospective study. *Gastroenterology* 1989;96:981–8.http://www.ncbi.nlm.nih.gov/pubmed/2494098 (accessed 6 Dec 2017).

56 Thomas MR, Litin SC, Osmon DR, *et al.* Lack of effect of Lactobacillus GG on antibiotic-associated diarrhea: a randomized, placebo-controlled trial. *Mayo Clin Proc* 2001;76:883–9. doi:10.4065/76.9.883

57 Velasco M, Requena T, Delgado-Iribarren A, *et al.* Probiotic Yogurt for the Prevention of Antibiotic-associated Diarrhea in Adults: A Randomized Double-blind Placebo-controlled Trial. *J Clin Gastroenterol* 2018;00:1–7. doi:10.1097/MCG.0000000000001131

58 Wenus C, Goll R, Loken EB, *et al.* Prevention of antibiotic-associated diarrhoea by a fermented probiotic milk drink. *Eur J Clin Nutr* 2008;62:299–301. doi:10.1038/sj.ejcn.1602718

59 Wong S, Jamous A, O'Driscoll J, *et al.* A Lactobacillus casei Shirota probiotic drink reduces antibiotic-associated diarrhoea in patients with spinal cord injuries: a randomised controlled trial. *Br J Nutr* 2013;111:672–8. doi:S0007114513002973 [pii]\r10.1017/S0007114513002973

60 Wright K, Wright H, Murray M. Probiotic treatment for the prevention of antibiotic-associated diarrhoea in geriatric patients: a multicentre randomised controlled pilot study. *Australas J Ageing* 2015;34:38–42. doi:10.1111/ajag.12116

61 Higgins J, Thomas J, Chandler J, *et al. Cochrane Handbook for Systematic Reviews of Interventions*. 2019. doi:10.1002/9781119536604

62 Dietrich CG, Kottmann T, Alavi M. Commercially available probiotic drinks containing Lactobacillus casei DN-114001 reduce antibiotic-associated diarrhea. *World J Gastroenterol* 2014;20:15837–44. doi:10.3748/wjg.v20.i42.15837

63 Price DD, Finniss DG, Benedetti F. A Comprehensive Review of the Placebo Effect: Recent Advances and Current Thought. *Annu Rev Psychol* 2008;59:565–90. doi:10.1146/annurev.psych.59.113006.095941

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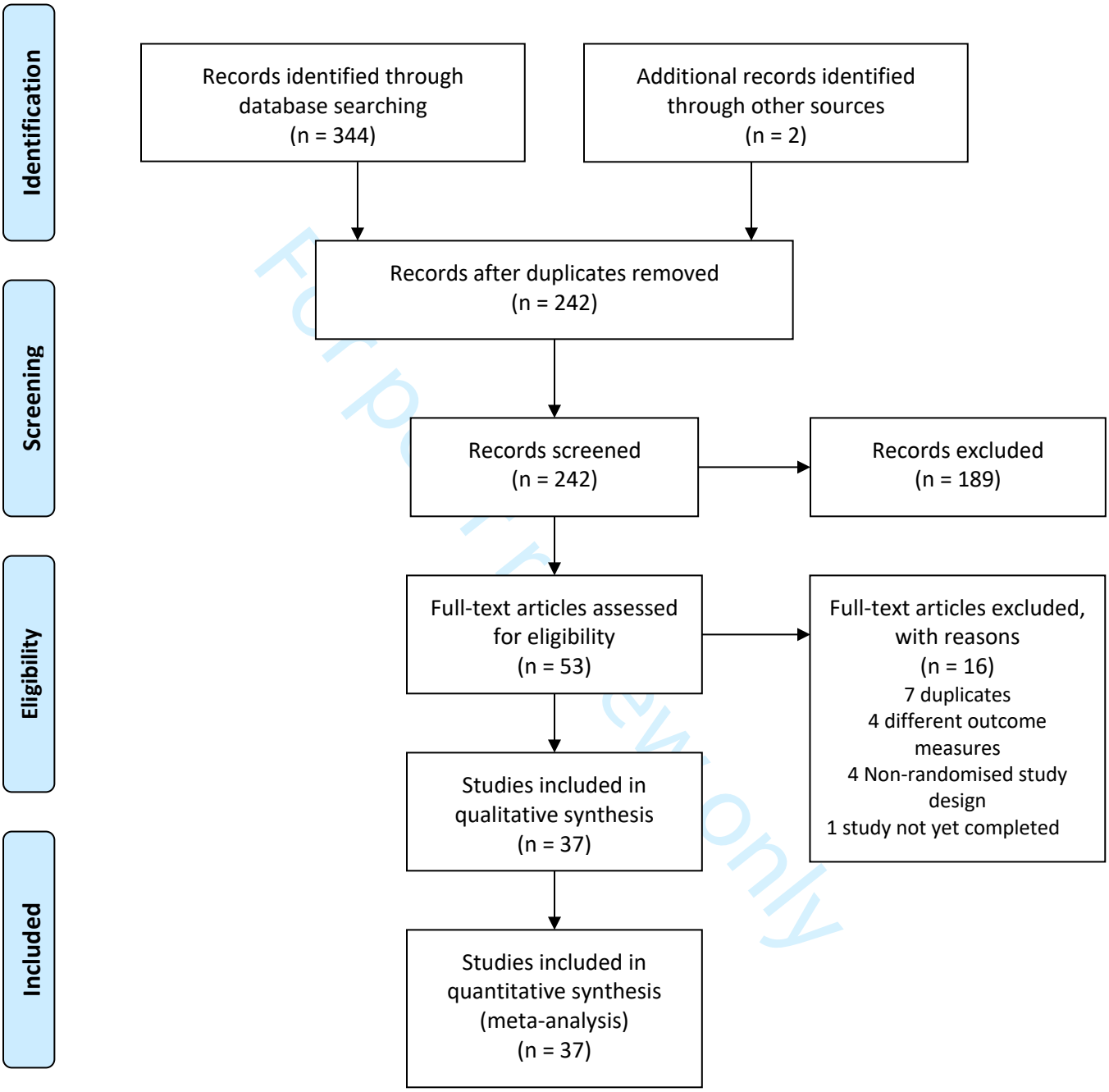
64 Hempel S, Newberry SJ, Maher AR, *et al.* Probiotics for the prevention and treatment of antibiotic-associated diarrhea: a systematic review and meta-analysis. *JAMA* 2012;307:1959–69. doi:10.1001/jama.2012.3507

65 Jafarnejad S, Shab-Bidar S, Speakman JR, *et al.* Probiotics Reduce the Risk of Antibiotic-Associated Diarrhea in Adults (18–64 Years) but Not the Elderly (>65 Years). *Nutr Clin Pract* 2016;31:502–13. doi:10.1177/0884533616639399

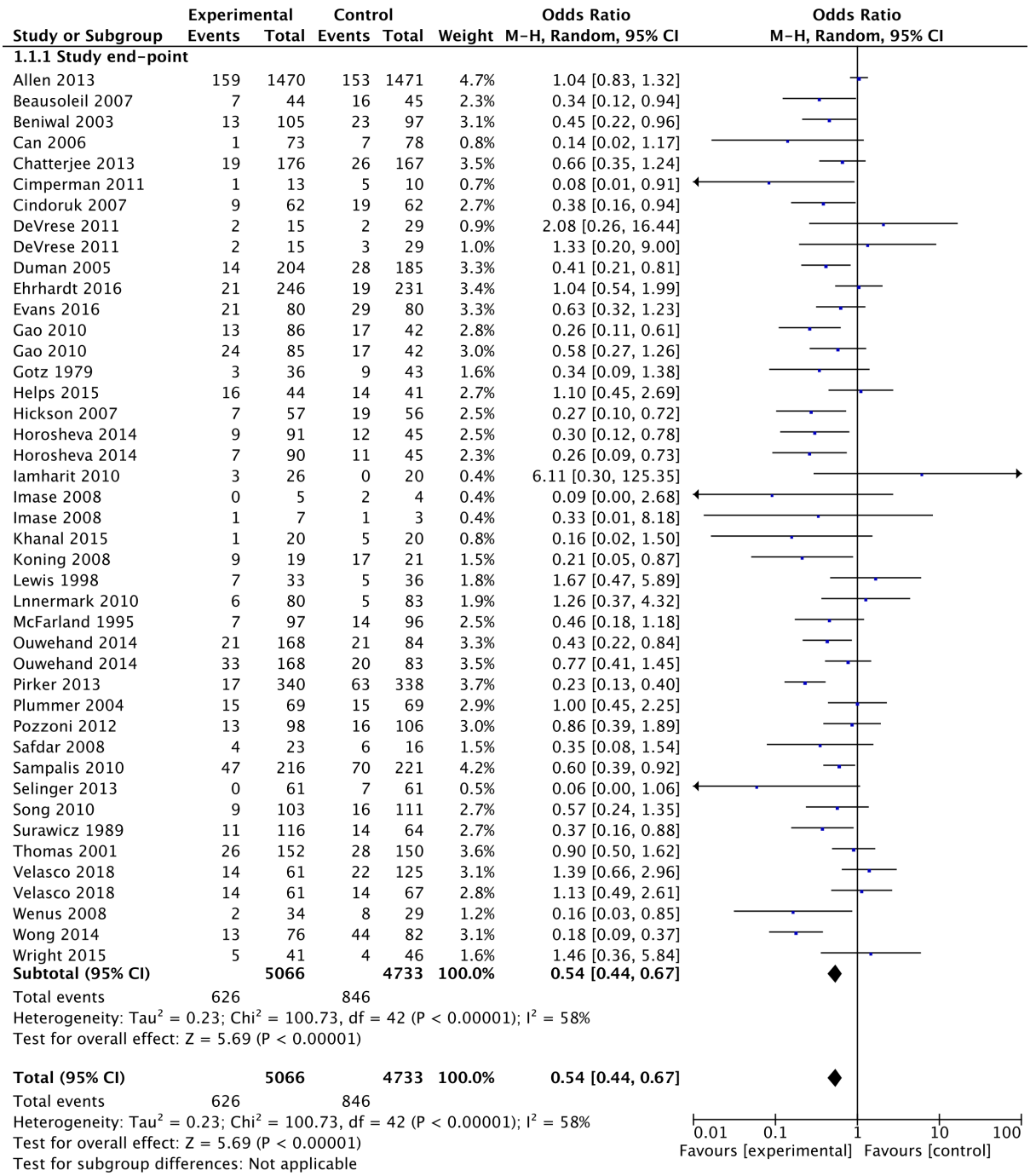
66 Xie C, Li J, Wang K, *et al.* Probiotics for the prevention of antibiotic- associated diarrhoea in older patients: A systematic review. *Travel Med Infect Dis* 2015;13:128–34. doi:10.1016/j.tmaid.2015.03.001

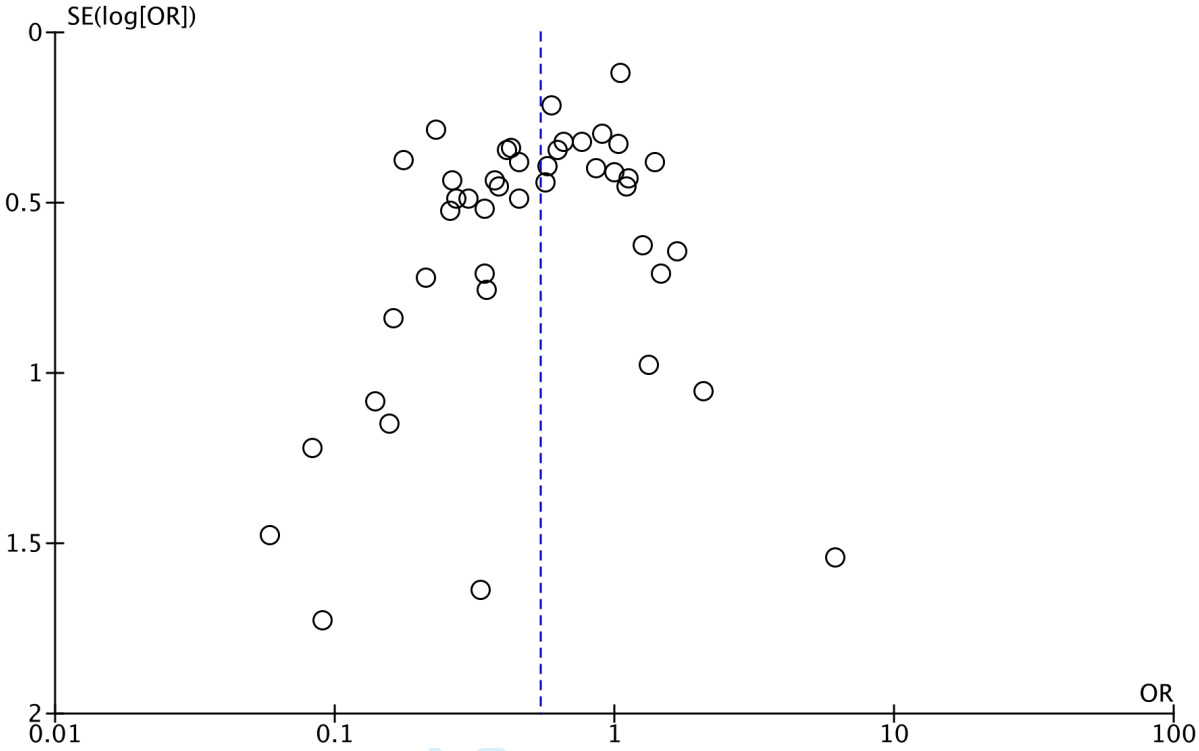
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Experimental
Control
Odds Ratio
Odds Ratio
Study or Subgroup Events Total Events Total Weight M-H, Random, 95% CI
M-H, Random, 95% CI
1.1.1 Study end-point
Allen 2013 159 1470 153 1471 4.7% 1.04 [0.83, 1.32]
Beausoleil 2007 7 44 16 45 2.3% 0.34 [0.12, 0.94]
Beniwal 2003 13 105 23 97 3.1% 0.45 [0.22, 0.96]
Can 2006 1 73 7 78 0.8% 0.14 [0.02, 1.17]
Chatterjee 2013 19 176 26 167 3.5% 0.66 [0.35, 1.24]
Cimperman 2011 1 13 5 10 0.7% 0.08 [0.01, 0.91]
Cindoruk 2007 9 62 19 62 2.7% 0.38 [0.16, 0.94]
DeVrese 2011 2 15 2 29 0.9% 2.08 [0.26, 16.44]
DeVrese 2011 2 15 3 29 1.0% 1.33 [0.20, 9.00]
Duman 2005 14 204 28 185 3.3% 0.41 [0.21, 0.81]
Ehrhardt 2016 21 246 19 231 3.4% 1.04 [0.54, 1.99]
Evans 2016 21 80 29 80 3.3% 0.63 [0.32, 1.23]
Gao 2010 13 86 17 42 2.8% 0.26 [0.11, 0.61]
Gao 2010 24 85 17 42 3.0% 0.58 [0.27, 1.26]
Gotz 1979 3 36 9 43 1.6% 0.34 [0.09, 1.38]
Helps 2015 16 44 14 41 2.7% 1.10 [0.45, 2.69]
Hickson 2007 7 57 19 56 2.5% 0.27 [0.10, 0.72]
Horosheva 2014 9 91 12 45 2.5% 0.30 [0.12, 0.78]
Horosheva 2014 7 90 11 45 2.3% 0.26 [0.09, 0.73]
Iamharit 2010 3 26 0 20 0.4% 6.11 [0.30, 125.35]
Imase 2008 0 5 2 4 0.4% 0.09 [0.00, 2.68]
Imase 2008 1 7 1 3 0.4% 0.33 [0.01, 8.18]
Khanal 2015 1 20 5 20 0.8% 0.16 [0.02, 1.50]
Koning 2008 9 19 17 21 1.5% 0.21 [0.05, 0.87]
Lewis 1998 7 33 5 36 1.8% 1.67 [0.47, 5.89]
Lnnermark 2010 6 80 5 83 1.9% 1.26 [0.37, 4.32]
McFarland 1995 7 97 14 96 2.5% 0.46 [0.18, 1.18]
Ouwehand 2014 21 168 21 84 3.3% 0.43 [0.22, 0.84]
Ouwehand 2014 33 168 20 83 3.5% 0.77 [0.41, 1.45]
Pirker 2013 17 340 63 338 3.7% 0.23 [0.13, 0.40]
Plummer 2004 15 69 15 69 2.9% 1.00 [0.45, 2.25]
Pozzoni 2012 13 98 16 106 3.0% 0.86 [0.39, 1.89]
Safdar 2008 4 23 6 16 1.5% 0.35 [0.08, 1.54]
Sampalis 2010 47 216 70 221 4.2% 0.60 [0.39, 0.92]
Selinger 2013 0 61 7 61 0.5% 0.06 [0.00, 1.06]
Song 2010 9 103 16 111 2.7% 0.57 [0.24, 1.35]
Surawicz 1989 11 116 14 64 2.7% 0.37 [0.16, 0.88]
Thomas 2001 26 152 28 150 3.6% 0.90 [0.50, 1.62]
Velasco 2018 14 61 22 125 3.1% 1.39 [0.66, 2.96]
Velasco 2018 14 61 14 67 2.8% 1.13 [0.49, 2.61]
Wenus 2008 2 34 8 29 1.2% 0.16 [0.03, 0.85]
Wong 2014 13 76 44 82 3.1% 0.18 [0.09, 0.37]
Wright 2015 5 41 4 46 1.6% 1.46 [0.36, 5.84]
Subtotal (95% CI) 5066 4733 100.0% 0.54 [0.44, 0.67]
Total events 626 846
Heterogeneity: Tau² = 0.23; Chi² = 100.73, df = 42 (P < 0.00001); I² = 58%
Test for overall effect: Z = 5.69 (P < 0.00001)
Total (95% CI) 5066 4733 100.0% 0.54 [0.44, 0.67]
Total events 626 846
Heterogeneity: Tau² = 0.23; Chi² = 100.73, df = 42 (P < 0.00001); I² = 58%
Test for overall effect: Z = 5.69 (P < 0.00001)
Test for subgroup differences: Not applicable
0.01 0.1 1 10 100
Favours [experimental] Favours [control]



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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	3
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	3
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	NA
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	4
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	4
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	4 / Appendix
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	4
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	4
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	4
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	4
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	4
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	4

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	4
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	4
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	5
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	5
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	6
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	7
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	7
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	8
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	8
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	8
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	9
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	10
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	11

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(6): e1000097. doi:10.1371/journal.pmed1000097

For more information, visit: www.prisma-statement.org.

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APPENDIX A: SEARCH STRATEGY FOR ELECTRONIC DATABASES

Databases: EMBASE, Medline (Ovid), Web of Science
Plus grey literature, hand searching, search of trial registries Australian New Zealand Clinical Trials Registry (ANZCTR) and ClinicalTrials.gov

P
Adults OR humans OR aged OR elderly OR young adults

I
Probiotic OR fermented milk OR fermented yoghurt OR lactic acid bacteria OR lactobacillus OR bifidobacterium OR saccharomyces OR lactococcus OR enterococcus OR streptococcus OR pediococcus OR leuconostoc OR bacillus OR esherichia OR coli OR rhamnosus OR acidophilus OR bifidum OR lactis OR bulgaricus OR reuteri OR casei OR thermophilus OR boulardii OR coagulans OR delbrueckii OR animalis OR thermophilus

C
Placebo OR no treatment OR pasteurised yoghurt OR heat-treated yoghurt OR control OR compar*

O
Incidence OR prevalence OR occurrence OR reduction OR improvement OR prevention OR alleviation OR episode* OR occasion

AND

Diarrh* OR antibiotic-associated diarrh* OR clostridium difficile associated diarrh* OR pseudomembranous colitis OR nosocomial diarrh* OR infectious diarrh*

S
Randomised controlled trial OR randomized controlled trial OR RCT

EMBASE and Medline (Ovid)

(Adults or humans or aged or elderly or young adults).m_titl.

(Probiotic or fermented milk or fermented yoghurt or lactic acid bacteria or lactobacillus or bifidobacterium or saccharomyces or lactococcus or enterococcus or streptococcus or pediococcus or leuconostoc or bacillus or esherichia or coli or rhamnosus or acidophilus or bifidum or lactis or bulgaricus or reuteri or casei or thermophilus or boulardii or coagulans or delbrueckii or animalis or thermophilus).m_titl.

(Placebo or no treatment or pasteurised yoghurt or heat-treated yoghurt or control or compar*).m_titl.

(Incidence or prevalence or occurrence or reduction or improvement or prevention or alleviation or episode* or occasion).m_titl.

(Diarrh* or antibiotic-associated diarrh* or clostridium difficile associated diarrh* or pseudomembranous colitis or nosocomial diarrh* or infectious diarrh*).m_titl.

(Randomised controlled trial or randomized controlled trial or RCT).m_titl.

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Web of Science

(#6 AND #5 AND #4 AND #3 AND #2 AND #1) *AND* **LANGUAGE:** (English) *AND* **DOCUMENT TYPES:** (Article OR Abstract of Published Item)
Refined by: DOCUMENT TYPES: (ARTICLE)
Indexes=SCI-EXPANDED, SSCI, A&HCI, CPCI-S, CPCI-SSH, ESCI, CCR-EXPANDED, IC
Timespan=All years

(#6 AND #5 AND #4 AND #3 AND #2 AND #1) *AND* **LANGUAGE:** (English) *AND* **DOCUMENT TYPES:** (Article OR Abstract of Published Item)
Indexes=SCI-EXPANDED, SSCI, A&HCI, CPCI-S, CPCI-SSH, ESCI, CCR-EXPANDED, IC
Timespan=All years

(TS=(Incidence or prevalence or occurrence or reduction or improvement or prevention or alleviation or episode* or occasion)) *AND* **LANGUAGE:** (English) *AND* **DOCUMENT TYPES:** (Article OR Abstract of Published Item)
Indexes=SCI-EXPANDED Timespan=All years

(TS=(Adults or humans or aged or elderly or young adults)) *AND* **LANGUAGE:** (English) *AND* **DOCUMENT TYPES:** (Article OR Abstract of Published Item)
Indexes=SCI-EXPANDED Timespan=All years

(TS=(Placebo or no treatment or pasteurised yoghurt or heat-treated yoghurt or control or compar*)) *AND* **LANGUAGE:** (English) *AND* **DOCUMENT TYPES:** (Article OR Abstract of Published Item)
Indexes=SCI-EXPANDED Timespan=All years

(TS=(Randomised controlled trial or randomized controlled trial or RCT)) *AND* **LANGUAGE:** (English) *AND* **DOCUMENT TYPES:** (Article OR Abstract of Published Item)
Indexes=SCI-EXPANDED Timespan=All years

TI=(Diarrh* or antibiotic-associated diarrh* or clostridium difficile associated diarrh* or pseudomembranous colitis or nosocomial diarrh* or infectious diarrh*)
Indexes=SCI-EXPANDED, SSCI, A&HCI, CPCI-S, CPCI-SSH, ESCI, CCR-EXPANDED, IC Timespan=All years

TI=(Probiotic or fermented milk or fermented yoghurt or lactic acid bacteria or lactobacillus or bifidobacterium or saccharomyces or lactococcus or enterococcus or streptococcus or pediococcus or leuconostoc or bacillus or esherichia or coli or rhamnosus or acidophilus or bifidum or lactis or bulgaricus or reuteri or casei or thermophilus or boulardii or coagulans or delbrueckii or animalis or thermophilus)
Indexes=SCI-EXPANDED, SSCI, A&HCI, CPCI-S, CPCI-SSH, ESCI, CCR-EXPANDED, IC Timespan=All years

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Probiotics for the prevention of antibiotic-associated diarrhoea: A systematic review and meta-analysis

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Primary Subject Heading:	Gastroenterology and hepatology
Secondary Subject Heading:	General practice / Family practice, Infectious diseases, Nutrition and metabolism, Pharmacology and therapeutics
Keywords:	BACTERIOLOGY, Gastrointestinal infections < GASTROENTEROLOGY, Gastroenterology < INTERNAL MEDICINE, PREVENTIVE MEDICINE, Adult gastroenterology < GASTROENTEROLOGY, NUTRITION & DIETETICS

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PROBIOTICS FOR THE PREVENTION OF ANTIBIOTIC-ASSOCIATED DIARRHOEA: A SYSTEMATIC REVIEW AND META-ANALYSIS

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ABSTRACT

Objective: To evaluate existing evidence for the use of probiotics in preventing antibiotic-associated diarrhoea (AAD) in adults.

Design: Systematic review and meta-analysis of randomised-controlled trials (RCTs).

Data sources: We performed a literature search of the electronic databases CINAHL Plus, EMBASE, MEDLINE (Ovid), and Web of Science from database inception to May 2021 as well as hand searching of trial registries and reference lists of related reviews.

Study selection: Two reviewers identified whether RCTs met the following inclusion criteria: adult population to whom antibiotics were administered; a probiotic intervention; a placebo, alternative dose, alternative probiotic strain or no treatment control; and incidence of AAD as the outcome.

Data extraction and synthesis: Two reviewers independently collected data and assessed risk of bias using pre-constructed data extraction forms. We used a random effects model for all analyses. Sub-group analyses were performed to evaluate species-specific and dose-specific response.

Results: Forty-two studies (11,305 participants) were included in this review. The pooled analysis suggests that co-administration of probiotics with antibiotics reduces the risk of AAD in adults by 37% (Risk ratio (RR) =0.63 [95% CI: 0.54, 0.73], p<0.00001). The overall quality of the evidence using the GRADE criteria was found to be moderate. In sub-group analyses, high dose compared with low dose of the same probiotic demonstrated a positive protective effect (RR 0.54 [95% CI: 0.38, 0.76], p<0.01), and only certain species, mainly of the lactobacillus and bifidobacteria genera, were found to be effective. Studies with a low baseline AAD risk did not show any difference in risk but studies with moderate or high baseline AAD risk demonstrated a significant risk reduction.

Conclusions: Probiotics are effective for preventing AAD. Secondary analyses of higher dosages and certain species have shown increased effectiveness. Our results may not be applicable in clinical scenarios of lower baseline AAD risk.

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STRENGTHS AND LIMITATIONS OF THIS STUDY

- An updated literature search encapsulating four years of new data since a comparable meta-analysis was last published by Cochrane in 2017
- We calculated and reported the number needed to be treated with probiotics to protect one person from antibiotic-associated diarrhoea (AAD), so that our data can be easily interpreted by patients and clinicians
- This meta-analysis looked only at incidence of AAD, but clinical applicability may have been improved by collecting data on duration and severity of AAD as well as length of stay in hospital and quality of life measures
- There was complexity in combining results due to substantial heterogeneity between study methodologies and outcome measures

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INTRODUCTION

Antibiotics are one of the most prescribed medications worldwide. They disturb normal gastrointestinal microbiota [1] and a common consequence is diarrhoea [2]. This leads to prolonged hospital admissions [3], increased morbidity and mortality [4] and greater costs to the health system [5,6]. Antibiotic associated diarrhoea (AAD) has been reported in 5-35% of patients receiving antibiotics, varying with the type of antibiotics, the general health of the patient and their exposure to pathogens [7].

Probiotics are live microorganisms that, when administered in adequate amounts, confer a health benefit on the host [8]. They are believed to improve the microbial balance of the host and reduce colonisation by pathogenic bacteria. They can be delivered orally, are thought to adhere to target gastrointestinal epithelium and are stable in acid and bile (Dunne et al. 2001). Probiotics are increasingly available in capsules, powders and fermented milk drinks and are promoted in health food shops and supermarkets, and in the media, for a multitude of purposes.

Several mechanisms are proposed to contribute to the protective effect of probiotics against AAD. Antibiotics are known to disrupt the complex balance of gastrointestinal microbiota, allowing colonisation by pathogenic organisms such as *Clostridium difficile* (*C. difficile*). Probiotics are believed to temporarily colonise the gut, producing bactericidal acids and peptides, and competing for nutrients and epithelial adhesion. Several species of the *Lactobacillus* genus are acid producing, lower environmental pH, and secrete exotoxins that inhibit pathogenic bacterial growth and inhibit binding of enterotoxins to gut epithelium [9–12]. Probiotics, particularly species in the *Lactobacillus* and *Saccharomyces* genera, are also reported to upregulate both the innate and adaptive immune systems through multiple mechanisms [13–16].

It is suggested that probiotics are the panacea to AAD. This would be a very attractive solution to an expensive problem with significant morbidity. It is estimated that the increase in length of hospital stay due to AAD is 3.6 to 21.3 days [17,18]. One of the most expensive probiotics in Australia costs less than nine dollars daily (VSL#3 probiotic formulation, AUD $8.60 per day, April 2021). This is negligible compared to the cost of one day of Australian hospital inpatient care, which according to 2017 data, cost on average $2128 [19].

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Our hypothesis has been extensively investigated. In a Cochrane review [20] authors found moderate quality evidence for a large protective effect of probiotics against *Clostridium difficile*-associated diarrhoea (CDAD) (RR 0.40, 95% CI 0.30 to 0.52). Although *C. difficile* is the most common infectious organism isolated, in most cases of AAD no causative agent is found. Faecal analysis indicates incidence of *C. difficile* in AAD to be 20% [21], or less [22]. Furthermore while probiotics significantly reduce incidence of CDAD they do not appear to reduce *C. difficile* infection [20]. Goldenberg and colleagues suggest that the role of probiotics may be more in alleviating the symptoms of infection rather than preventing colonisation itself [20]. For this reason, we examined the protective effect of probiotics on the occurrence of AAD without any specific causative organism.

The primary objective of this systematic review and meta-analysis was to evaluate existing evidence for the protective effect of probiotics against AAD in adults. Secondary objectives were to examine dose-response and species-specific response in preventing AAD.

METHODS

Eligibility criteria

Randomised controlled trials (RCTs) comparing probiotic use (any strain, dose or formulation) to placebo, alternative dose, alternative probiotic strain, or no treatment, for the prevention of diarrhoea in adults receiving antibiotic therapy, were eligible for inclusion in this review. Studies of adults, receiving any antibiotic therapy for a course of any length, or for any reason, were included. Studies of paediatric populations were excluded. The primary outcome measure was incidence of AAD, as defined by authors, during the antibiotic treatment or follow-up phases. Studies were excluded if they examined probiotics only for treatment, rather than prevention, of AAD. We limited included studies to those written in English as our group did not speak any other languages.

Information sources and search strategy

A comprehensive electronic title search (Appendix A) was performed of CINAHL Plus, EMBASE, MEDLINE (Ovid), Web of Science and Google Scholar from database inception to May 2021. The search was broad to include as many individual probiotic species as possible and all clinical indications or outcomes (full search terms in Appendix A). Grey literature and clinical trial registers including ClinicalTrials.gov and the Australia New Zealand Clinical Trials Registry (ANZCTR) were searched, as well as hand searching of reference lists of relevant

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systematic reviews and included studies. The search was limited to RCTs in an adult population.

Study selection

Two of three of the authors (CG, GK and KL) independently assessed each study for inclusion in the review. Discrepancies were resolved through discussion.

Data collection process and data items

Data were extracted from each article by two of three of the authors (CG, GK, KL), using a standardised data extraction form following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines[23]. This was housed within Covidence online software (Covidence systematic review software, Veritas Health Innovation, Melbourne, Australia. Available at www.covidence.org). Data items collected included the number of patients allocated to each group, the probiotic species, formulation, dose and regime, whether a placebo or 'nothing' control was used, and outcome data on incidence of AAD.

Assessment of risk of bias in individual studies and overall quality of the evidence

The risk of bias was assessed using the Cochrane Risk of Bias [24] tool within the Covidence online software. The domains assessed were sequence generation, allocation concealment, blinding of participants and personnel, blinding of outcome assessors, incomplete outcome data, selective outcome reporting and other sources of bias (e.g. differences in baseline characteristics or funding of trials by pharmaceutical companies). Assessment was conducted by two of three authors (CG, GK, KL) and discrepancies resolved by consensus. The overall quality of the evidence was rated using the GRADE approach [25] independently by two authors (CG, KL) and consensus reached through discussion.

Statistical analysis and measures of treatment effect

For data analysis we used the RevMan 5.4.1 statistical software (Review Manager (RevMan) [Computer program]. Version 5.4. The Cochrane Collaboration, 2020.) Using a random-effects model, dichotomous data were reported as a risk ratio (RR) with corresponding 95% confidence intervals (95% CI). The number needed to treat (NNT) was also calculated for the complete case analysis. The I^2 statistic was reported for statistical heterogeneity, although

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clinical and methodological heterogeneity would not have been reflected within this test. Our *a priori* clinical heterogeneity variables included: probiotic dose, probiotic species and risk of bias. We also conducted *post hoc* sub-group analyses on baseline risk for our primary outcome, incidence of AAD, and concurrent use of probiotics with H. pylori eradication. Sensitivity analyses were performed where deemed necessary.

Patient and public involvement

There was no patient or public involvement in this study.

RESULTS

Study selection and characteristics

A total of 356 articles were identified in the initial search, of which 60 were eligible for full-text review, and 42 of those articles (11,603 participants) were eligible for inclusion in the study. Five of these were published after the Cochrane review mentioned previously [20]. Figure 1 summarises the PRISMA flow of studies. The characteristics of each included study are presented in table 1.

Table 1: Characteristics of included studies

Reference	Total participants (N probiotic: N control)	Health care setting & age of patients	Probiotic species	Daily probiotic dose (cfu)	Duration of probiotic therapy	Control	Outcome (AAD) measurement
Allen, 2013 [26]	2941 (1470:1471)	Inpatient ≥65	L. acidophilus, B. bifidum	60 x 10^9	21 days	Placebo	≥3 loose stools (5-7 Bristol Stool Scale) in 24 hours
Beausoleil, 2007 [27]	89 (44:45)	Inpatient 54-85	L. acidophilus, L. casei (Bio-K+ CL1285, Bio-K+ International Inc, Canada)	50 x 10^9	Duration of antibiotic	Placebo	≥3 loose stools (5-7 Bristol Stool Scale) in 24 hours
Beniwal, 2003 [28]	202 (105:97)	Inpatient >18	L. acidophilus, L. bulgaricus, S. thermophilus	0.002 x 10^9	8 days	Nothing	>2 loose stools per day
Bravo, 2008 [29]	86 (41:45)	Outpatient >15	S. boulardii	10.2 x 10^9	12 days	Placebo	≥3 loose stools per day for ≥2 days
Can, 2006 [30]	151 (73:78)	Inpatient 25-50	S. boulardii	Not noted	Not noted	Placebo	Not noted
Chatterjee, 2013 [31]	343 (176:167)	Outpatient >18	L. acidophilus LA-5®, B. animalis ssp. lactis BB-12®	4 x 10^9	Duration of antibiotic	Placebo	≥3 loose stools (5-7 Bristol Stool Scale) for ≥2 days
Cimperman, 2011 [32]	23 (13:10)	Inpatient >18	L. reuteri ATCC 55730 (BioGaia Biologics, Sweden)	0.2 x 10^9	28 days	Placebo	≥3 loose stools (5-7 Bristol Stool Scale) for ≥2 days

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Cindoruk, 2007 [33]	124 (62:62)	Inpatients being treated for H. pylori	S. boulardii	1g (cfu not noted)	14 days	Placebo	Not noted
De Vrese, 2011 [34]	88 (30:29:29) (Probiotic: pasteurised: acidified)	Outpatients being treated for H. pylori	L. acidophilus, LA-5®, B. animalis, ssp. lactis, BB-12®, S. thermophilus	0.25 x 10^9	56 days	Pasteurised and acidified placebos	≥3 watery stools in one day
Duman, 2005 [35]	389 (204:185)	Outpatients being treated for H. pylori	S. boulardii	1g (cfu not noted)	14 days	Nothing	≥3 loose stools per day for ≥2 consecutive days
Ehrhardt, 2016 [36]	477 (246:231)	Inpatient 40-77	S. boulardii	36 x 10^9	Duration of antibiotic + 7 days	Placebo	≥3 loose stools (5-7 Bristol Stool Scale) in 24 hours
Evans, 2016 [37]	160 (80:80)	Healthy outpatients 18-50	L. helveticus + L. rhamnosus	4 x 10^9	Duration of antibiotic + 7 days	Placebo	≥3 loose stools (5-7 Bristol Stool Scale) in 24 hours
Gao, 2010 [38]	255 (86:85:84) (High: low: placebo)	Inpatient 50-70	L acidophilus CL1285®, L. casei LBC80R®	100 x 10^9, 50 x 10^9	Duration of antibiotic + 7 days	Placebo	≥3 loose stools (5-7 Bristol Stool Scale) in 24 hours
Gotz, 1979 [39]	79 (36:43)	Inpatient >18	L. acidophilus (gasseri), L. helveticus (bulgaricus) (Lactinex)	4 x 10^9	5 days	Placebo	≥3 stools more than normal number of evacuations in 24 hours
Helps, 2015 [40]	85 (44:41)	Inpatient >18	L. casei Shirota (Yakult)	6.5 x 10^9	Duration of antibiotic + 7 days	Placebo	≥3 loose stools in 24 hours
Hickson, 2007 [41]	113 (57:56)	Inpatient >50	L. casei defensis, L. bulgaricus, S. thermophilus (Actimel)	0.21 x 10^9	Duration of antibiotic +5 days	Placebo	≥3 loose stools (5-7 Bristol Stool Scale) in 24 hours
Horosheva, 2014 [22]	271 (91:90:90) (Species1: species 2: placebo)	Outpatient ≥45	(1) B. subtilis 3 & B. licheniformis 31, (2) B. licheniformis only	4 x 10^9	Duration of antibiotic + 7 days	Placebo	≥3 loose stools (5-7 Bristol Stool Scale) for ≥2 days
Iamharit, 2010 [42]	46 (26:20)	Inpatient	Lyophilized live L. acidophilus + B. bifidum (Infloran®)	6 x 10^9	14 days	Placebo	≥3 loose stools per day for ≥2 consecutive days or ≥5 loose stools in a day
Imase, 2008 [43]	19 (5:7:7) (High: low: nothing)	Inpatients being treated for H. pylori	C. Butyricum (CMB588)	0.02 x 10^9, 0.01 x 10^9	7 days	Nothing	Loose or mostly loose stools
Khanal, 2015 [44]	40 (20:20)	Outpatients undergoing treatment for H. pylori	S. boulardii, Lactobacillus and B. longum	5 x 10^9	14 days	Nothing	≥3 loose stools per day
Koning, 2008 [45]	38 (19:19)	Healthy 18-65	B. bifidum W23, B. lactis W18, B. longum W51, E. faecium W54, L. acidophilus W37 and W55, L. paracasei W72, L. plantarum W62, L. rhamnosus W71, L. salivarius W24	2 x 10^9	14 days	Placebo	≥3 loose stools (5-7 Bristol Stool Scale) in 24 hours
Lewis, 1998 [46]	69 (33:36)	Inpatient >18	S. boulardii	Not noted	Duration of antibiotic	Placebo	≥3 loose stools (on simplified Bristol Stool Scale) in 24 hours

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Lönnermark, 2010 [47]	163 (80:83)	Inpatient >16	L. plantarum 299v	10 x 10^9	Duration of antibiotic + 7 days	Placebo	≥3 loose stools (5-7 Bristol Stool Scale) for ≥2 days
McFarland, 1995 [48]	193 (97:96)	Inpatient >18	S. boulardii	Not noted	Duration of antibiotic + 3 days	Placebo	≥3 loose stools (5-7 Bristol Stool Scale) for ≥2 days
Ouwehand, 2014 [49]	503 (168:168:167) (High: low: placebo)	Inpatient 30-70	L. acidophilus NCFM® (ATCC700396), L. paracasei Lpc-37 (ATCC SD5275), B. lactis Bi-07 (ATCC SD5220), B. lactis BI-04 (ATCC SD5219)	17 x 10^9, 4.17 x 10^9	Duration of antibiotic + 7 days	Placebo	≥3 loose stools (5-7 Bristol Stool Scale) in 24 hours
Pirker, 2013 [50]	678 (340:338)	Inpatient >18	L. casei Shirota (Yakult)	6.5 x 10^9	Duration of antibiotic + 3 days	Nothing	≥3 loose stools per day for ≥3 days
Plomer, 2020 [51]	130 (65:65)	Outpatients 18-65 being treated for H. pylori	Bacillus. clausii	6 x 10^9	Duration of antibiotic + 7 days	Placebo	Not noted
Plummer, 2004 [52]	20 (69:69)	Inpatient >18	L. acidophilus, B. bifidum	20 x 10^9	20 days	Placebo	Not noted
Pozzoni, 2012 [53]	204 (106:98)	Inpatient >50	S. boulardii	10 x 10^9	Duration of antibiotic +7 days	Placebo	≥3 loose stools (5-7 Bristol Stool Scale) for ≥2 days
Rajkumar, 2020 [54]	1126 (549:577)	Inpatient ≥55	L. casei DN114001, L. delbrueckii subspecies bulgaricus, S. thermophilus	20.4 x 10^9	Duration of antibiotic + 7 days	Placebo	≥2 loose stools (5-7 Bristol Stool Scale) in 24 hours
Safdar, 2008 [55]	39 (23:16)	Inpatient >18	L. acidophilus	60 x 10^9	Duration of antibiotic + 14 days	Placebo	≥2 consecutive days of Stool Consistency Continuum score 1, 2 or 3
Sampalis, 2010 [56]	437 (216:221)	Inpatient >18	L. acidophilus + L. casei (Bio-K+ CL1285, Bio-K+ International Inc, Canada)	50 x 10^9	Duration of antibiotic + 5 days	Placebo	≥1 day of ≥1 episode of loose stool
Selinger, 2013 [57]	122 (61:61)	Inpatient >18	B. breve, B. longum, B. infants, L. acidophilus, L. plantarum, L. paracasei, L. delbrueckii subsp. bulgaricus, S. thermophilus (VSL#3)	900 x 10^9	Duration of antibiotic + 7 days	Placebo	≥3 loose stools (5-7 Bristol Stool Scale) in 24 hours
Shimbo, 2005 [58]	35 (18:17)	Outpatients >18 being treated for H. pylori	C. butyricum MIYAIRI 588	Not noted	14 days	Nothing	Not noted
Song, 2010 [59]	214 (103:111)	Inpatient >18	L. rhamnosus R0011 + L. acidophilus R0052	4 x 10^9	14 days	Placebo	≥3 loose stools (5-7 Bristol Stool Scale) in 24 hours
Surawicz, 1989 [60]	180 (116:64)	Inpatient >18	S. boulardii	Not noted	Duration of antibiotic + 14 days	Placebo	≥3 loose or watery stools per day for ≥2 consecutive days
Thomas, 2001 [61]	267 (133:134)	Inpatient >18	L. rhamnosus GG	20 x 10^9	14 days	Placebo	Loose stools (1-3 on Stool Consistency Continuum) for ≥2 or more consecutive days,

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							or ≥3 more than patient's normal daily number of bowel movements
Velasco, 2019 [62]	314 (122:125:67) (Probiotic: placebo: nothing)	Inpatient >18	L. acidophilus La-5, L. casei Lc-01, B. lactis Bb-12 as probiotics and S. thermophilus STY-31, L. bulgaricus LBY-27 as starter strains in both yoghurt drinks	17-23 x 10^9 plus starter strains	Duration of antibiotic + 5 days	Placebo and nothing control groups	≥3 loose stools (5-7 Bristol Stool Scale) for ≥2 days
Wenus, 2008 [63]	63 (34:29)	Inpatient >18	L. rhamnosus GG (LGG), L. acidophilus La-5 and B. animalis subsp. lactis BB-12®	52.5 x 10^9	Not noted	Placebo	≥3 loose stools per day for consecutive 2 days during the test period
Wong, 2013 [64]	158 (76:82)	Inpatient >18	L. casei Shirota (Yakult)	6.5 x 10^9	Duration of antibiotic + 7 days	Nothing	≥3 loose stools (5-7 Bristol Stool Scale) for ≥2 days
Wright, 2015 [65]	86 (41:46)	Inpatient >65	L. casei Shirota (Yakult)	13 x 10^9	Duration of admission	Placebo	≥3 loose stools (5-7 Bristol Stool Scale) for ≥2 days
Wu, 2020 [66]	196 (65:67:64) (High: low: nothing)	Inpatient >18	B. longum, L. acidophilus, E. faecalis	48 x 10^9 or 24 x 10^9	14 days	Nothing	Not noted

All included studies were RCTs, conducted in adult populations. The overall age range was from 15 to 85 years. Interventions were comparable, in terms of all being probiotics, however, there was variation in the dose and species of probiotic used as well as the probiotic formulation i.e. tablets, powder, yoghurt or fermented milk drink. We investigated these by sub-group analysis. All included studies reported on the outcome of interest; incidence of AAD, however, this was measured differently between studies and was not the primary research question of the trial in all cases.

Risk of bias within studies

The risk of bias summary is displayed in figure 2. Six of the 42 studies were rated as having a low risk of bias, while 36 were rated as having a high or unclear risk of bias. On sensitivity analysis, the low risk of bias studies demonstrated no protective effect of probiotics (RR=0.78, [95% CI 0.57, 1.07], p=0.13) as opposed to the high or unknown risk of bias studies (RR 0.59, [95% CI 0.50 to 0.70], p<0.00001).

Results of individual studies and synthesis of results

Of the 42 studies included (table 1), 33 were placebo-controlled and eight provided 'no treatment' as the comparison. One study compared their treatment group with both a placebo group and a no treatment control [62]. Four trials compared two probiotic arms of differing doses with a control group [38,43,49,66]. Where there were two treatment groups (e.g. high

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dose and low dose) and one control group, the control group was split so that half served as comparator for each arm, in accordance with the Cochrane Systematic Review Guidelines [67]. Similarly in one study [34] where a probiotic group was compared with a placebo and 'no treatment' control, the probiotic group was split. In one study [62] the placebo milk contained a higher concentration of starter strains than the probiotic milk. We combined the groups to make it an overall treatment group, which the authors had done themselves in the 'both' group, compared with a 'no treatment' group. Only one study was found to compare two strains of the same probiotic species, however, it was excluded as it was not randomised [68].

The overall pooled results of the complete case analysis of 42 studies (reporting on 11,305 participants) (Figure 3), comparing probiotics with any control, favoured probiotics demonstrating a statistically significant reduction in the incidence of AAD. The pooled risk ratio of 0.63 (RR=0.63 [95% CI: 0.54, 0.73], p<0.00001) indicates that those who took probiotics were over one third less likely to develop AAD. This data suggests that the NNT, to prevent one case of AAD would be 20 patients. Statistically significant heterogeneity was detected for this comparison (P<0.00001; I^2=60%). We used a random effects model for this analysis based on the heterogeneity of the study methodologies.

Sub-group and heterogeneity analysis

When sub-group analyses were performed, where probiotics were compared to placebo (RR=0.69 [95% CI: 0.60, 0.80], p=0.0004), and where probiotics were compared to no treatment (RR=0.35 [95% CI: 0.27, 0.47], p<0.00001), a significant reduction in relative risk of diarrhoea was noted (Forest plots available in supplemental files). The difference in effect size may be explained as placebo effect.

Across all of the sub-group analyses, the I^2 statistic ranged from zero to 78%. There was also substantial clinical and methodological heterogeneity, therefore we used a random effects model for all analyses. There was substantial variation in the AAD definition employed in the included studies (Table 1). Although 13 studies defined AAD as ≥3 loose stools (5-7 Bristol Stool Scale) in 24 hours, 12 defined AAD as ≥3 loose stools (5-7 Bristol Stool Scale) in two days or more, 11 studies used various other definitions of AAD and five studies made no mention of their definition.

Sub-group analysis: probiotic dose

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Four studies directly compared high dose and low dose of the same probiotic [38,43,49,66]. The higher dose group demonstrated a significant reduction in the relative risk of AAD (RR 0.54 [95% CI: 0.38, 0.76], p=0.0004) (Forest plot available in supplemental files).

Sub-group analysis: probiotic species

Most studies used probiotic formulations containing more than one probiotic species so a sub-group analysis was performed on all individual species mentioned in included studies (Table 2). Where only one study existed that used a particular strain, we analysed that study's outcome data alone. The following probiotic strains demonstrated significant reduction in AAD when compared to control: *L. acidophilus, L. bulgaricus, L. casei, L. paracasei, L. rhamnosus, Lactobacillus spp., S. boulardii, B. animalis ssp. Lactis, B. longum, B. licheniformis, B. subtilis and Bacillus clausii.* However, L. helveticus, *L. plantarum, L. reuteri, B. bifidum, S. thermophilus* and *C. butyricum* provided no improvement in AAD compared to control.

Table 2: Sub-group analyses by species or genus (Forest plots for all sub-group analyses available in supplemental files)

Species	*Studies* (N)	*Effect size*
L. acidophilus	18 (6077)	RR=0.66 [95% CI: 0.54, 0.80], p<0.0001
L. bulgaricus	7 (1996)	RR=0.60 [95% CI: 0.37, 0.97], p=0.04
L. casei	11 (3382)	RR=0.59 [95% CI: 0.42, 0.83], p=0.003
L. helveticus	2 (239)	Not significant
L. paracasei	3 (665)	RR=0.60 [95% CI: 0.41, 0.88], p=0.01
L. plantarum	3 (325)	Not significant
L. reuteri	1 (23)	Not significant
L. rhamnosus	5 (779)	RR=0.71 [95% CI: 0.52, 0.97], p=0.03
Lactobacillus spp.	28 (7851)	RR=0.63 [95% CI: 0.52, 0.76], p<0.00001
S. boulardii	9 (1827)	RR=0.63 [95% CI: 0.46, 0.86], p=0.004
B. animalis ssp. lactis	6 (1351)	RR=0.70 [95% CI: 0.54, 0.91], p=0.009
B. bifidum	4 (3165)	Not significant
B. licheniformis	1 (181)	RR=0.39 [95% CI: 0.19, 0.79], p=0.009

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B. longum	4 (366)	RR=0.46 [95% CI: 0.28, 0.73], p=0.001
B. subtilis	1 (271)	RR=0.35 [95% CI: 0.19, 0.62], p=0.0004
S. thermophilus	6 (1917)	Not significant
C. butyricum	1 (19)	Not significant
Bacillus clausii	1 (130)	RR=0.61 [95% CI: 0.41, 0.89], p=0.01

Sub-group analysis: Concurrent *H. pylori* treatment

In the seven RCTs investigating adjunct probiotic use with antibiotics for *H. pylori* treatment we found a reduction in diarrhoeal side effects of 45% (RR=0.55, [95% CI: 0.41, 0.73], p<0.0001) (Forest plot available in supplemental files).

Sensitivity analysis: Baseline risk of AAD

In a *post hoc* analysis we divided our included studies into low (0-10%; 7 studies), moderate (11-30%; 21 studies) and high (>31%; 14 studies) baseline risk of AAD. The pooled studies with a low baseline risk of AAD did not show any difference in risk, but trials enrolling participants with a moderate or high baseline risk for developing AAD demonstrated 39% (RR=0.61, [95% CI: 0.48, 0.78], p<0.0001) and 45% (RR=0.55, [95% CI: 0.46, 0.66], p<0.00001) risk reduction respectively (Forest plots available in supplemental files).

Sensitivity analysis: High/unknown risk of bias

To examine for robustness we pooled data from the studies which we assessed as low risk of bias in all domains, and separately pooled the studies which scored high or unknown risk of bias in any domain. In our sensitivity analysis with the high and unknown risk of bias studies removed, the protective effect of probiotics against AAD was no longer statistically significant (RR=0.78 [95% CI: 0.57, 1.07], P =0.13). However, with only the high risk of bias studies removed from the complete data set, a similar pooled protective effect of probiotics was found (RR=0.65 [95% CI: 0.54, 0.78], p<0.00001) (Forest plots available in supplemental files).

Adverse events

Of the 42 included studies, 32 reported on adverse events. No serious adverse events were reported in any of the included studies. One study reported four (2.0%) non-serious adverse events in the probiotic group compared with none in the control group [30] and another

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commented that their probiotic milk was less well tolerated than their placebo milk (Velasco 2019).

Small study effects

On inspection, the funnel plot of the complete case analysis (Figure 4) is symmetrical which suggests little evidence of publication bias.

Overall quality of evidence

Using the GRADE approach [25] we rated the overall quality of the evidence as moderate. Although all included studies were RCTs, many demonstrated design limitations, mainly reflecting the reporting of sequence generation and allocation concealment in older studies, which suggests some risk of bias. This was balanced, however, by a good magnitude of effect and dose-response gradient, increasing the certainty of this body of evidence.

DISCUSSION

Summary of evidence

The principal finding of this review is that probiotics significantly reduce the risk of AAD in adults, with an NNT of 20 patients. These results were robust in sub-group analyses, with high doses more effective than lower doses, and some species more effective than others. As is the nature of systematic reviews, we found significant heterogeneity in study designs and population, severity of illness of the study group, antibiotics administered, as well as probiotic species and strains used, dose, timing and duration of probiotic administration. It may be important to note that our sub-group analysis of low baseline risk studies failed to find a statistically significant effect of probiotics. Therefore our overall effect estimate may not be directly applicable to clinical scenarios with inherent lower baseline risk of AAD. Using the GRADE criteria [25], the quality of the evidence included in our meta-analysis was moderate and therefore it would be appropriate to base future clinical practice on our results.

Our results show evidence of a placebo effect, with a reduction in risk of AAD of 31% when probiotics were compared to placebo, but 65% when probiotics were compared to no treatment. Placebo effect is not well understood but this could be attributed to expectations of symptom change with neurobiological underpinnings and actual effects on the brain and body [69]. This effect could be investigated further in well-designed, three-armed, RCTs.

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Secondary objectives included sub-group analyses of dose-response and species-specific response. In studies where dosages of the same species were compared [38,43,49,66], higher doses showed a significantly greater benefit (RR=0.54). It is difficult to interpret the dosage effect as the four studies that compared different doses of the same probiotic species made comparisons of differing magnitudes. Probiotic course duration was reported in most studies (40 out of 42), ranging from five to 56 days. Most commonly, probiotics were administered for the duration of antibiotic therapy plus a further seven days, or a total of 14 days. It would be biologically plausible to suggest that probiotic use should be continued for some period after the cessation of antibiotics. This is another question worth investigating in future trials.

Many studies used blends of probiotic species and strains, which made it problematic to establish independent associations. In our sub-group analyses, the individual probiotic species that demonstrated the greatest effect sizes were *B. licheniformis* (1 study, RR=0.39), *B. longum* (4 studies, RR=0.46) and *B. subtilis* (1 study, RR=0.35). Although demonstrating a more moderate effect size, *L. acidophilus* (18 studies, RR=0.66), *L. casei* (11 studies, RR=0.59) and *S. boulardii* (9 studies, RR=0.63) were used in many more studies. Overall, the lactobacillus genus showed a good effect with a small confidence interval. The individual species *L. acidophilus* was used in the greatest number of studies, demonstrating a moderate effect with a relatively small confidence interval. Moderate effect sizes were noted with other species and strains, which may be worth looking into more specifically in future trials. *L. helveticus, L. plantarum*, *L. reuteri, B. bifidum*, *S. thermophilus* and *C. butyricum* did not show evidence of effectiveness in reducing incidence of AAD.

We found only one study [68] comparing two strains of the same probiotic species, *L. casei DN-114001,* (Actimel®) and *L. casei Shirota* (Yakult). In this study *L. casei DN-114001* was significantly more effective for preventing AAD than *L. casei Shirota*. However, we were unable to include this data in our review as it was not randomised. Future RCTs should focus on direct comparisons of single or combined probiotic species and strains.

The most commonly identified indication for antibiotics was treatment of *H. pylori*. The reduction in diarrhoeal side effects by 45%, through the use of adjunct probiotics, may be a significant finding in improving quality of life for these patients.

Limitations

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A limitation of this study is that the only outcome measured was incidence of diarrhoea. This review's utility in clinical practice may have been improved by also analysing duration of AAD, diarrhoea severity, need for hospitalisation due to diarrhoea and quality of life measures. Additionally, it may have been informative to correlate outcomes with the duration of probiotic course or the date of probiotic initiation in relation to antibiotics.

Sub-group analysis by antibiotic would have been informative but was not possible as many of the included studies did not state the antibiotic name, class, route of administration, dose or course length that participants were administered. Future studies could focus on this.

This study was somewhat limited in scope as we only included studies published in English. Additionally, the initial search was a title search as opposed to keyword search, which may explain the lower number of studies initially identified, compared to the recent Cochrane review [20].

Agreements and disagreements with other studies

This study is concordant with findings from the aforementioned Cochrane systematic review [20]. Goldenberg found that probiotics led to a significant reduction in incidence of AAD from 18% to 12% of participants (R 0.58, 95% CI 0.48 to 0.70). The present review included 18 additional studies (4487 additional participants). This increase in total population increases the power of our review to detect smaller clinically relevant differences between groups in incidence of AAD.

Our sub-group analyses found similarly to one previous meta-analysis [70] that *Lactobacillus* and *Saccharomyces* genera reduced incidence of AAD. A more recent meta-analysis [71] found, also concordantly, that many *Lactobacillus* and *Bifidobacteria* species and *S. boulardii* were effective. Previous meta-analyses have not reported on dose-response.

Our sub-group analyses were also in agreement with The American Gastroenerological Association Clinical Practice Guidelines on the Role of Probiotics in the Management of Gastrointestinal Disorders [72] which suggest the use of S. boulardii; or the 2-strain combination of L. acidophilus CL1285 and L. casei LBC80R among other probiotic species for prevention of AAD.

Contrary to the findings of our review, one meta-analysis [73] reported that probiotics may not be effective in reducing the incidence of AAD, however, only 13 studies were analysed,

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including adults over 65 only. This suggests a need for looking separately at the effectiveness of probiotics in populations of younger and older adults.

CONCLUSIONS

This review found evidence supporting a moderate protective effect for probiotics in preventing AAD. In absolute terms, the NNT, to prevent one case of AAD would be 20 patients. These results should be applied with caution in clinical scenarios with inherent lower baseline risk of AAD. In studies that compared dosage of the same probiotic higher doses were found to be significantly more effective, suggesting a recommended dose of over 10 billion cfu per day. Although the probiotic species that demonstrated the greatest effect were *B. licheniformis*, *B. longum*, and *B. subtilis,* each was only studied in a handful of trials. The species *L. acidophilus*, *L. casei* and *S. boulardii*, still demonstrating moderate effect sizes in preventing AAD, were used in many more trials. A good effect size was found supporting the use of probiotics, given concurrently with antibiotics for H. pylori eradication therapy.

This review has demonstrated overall effectiveness of probiotics for the prevention of AAD, including a dose-response and some species-specific effectiveness of probiotics. Further high-quality trials are needed, focusing on direct comparison of doses, comparison of individual species and combinations of species. Future studies should record not only incidence of diarrhoea, but also duration, severity, quality of life measures and length of stay in hospital so that an assessment of the cost-benefit of probiotic therapy can be made. Future trials should ensure there is adequate blinding and reporting of adverse events.

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CONTRIBUTORSHIP STATEMENT

Study concept and design: CG, CH, KL
Acquisition of data: CG, GK, KL
Analysis and interpretation of data: CG, GK, KL
Drafting of the manuscript: CG, KL
Critical revision of the manuscript for important intellectual content: CG, GK, EG, KL
Statistical analysis: CG, EG, KL
Study supervision: KL

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COMPETING INTERESTS

None to declare

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ETHICS APPROVAL

This systematic review and meta-analysis was based on published data. As researchers did not access any information that could lead to the identification of an individual patient, no concerning ethical issue was raised in this research. Therefore, obtaining ethical approval and consent of participants was waived.

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DATA SHARING STATEMENT

Raw data used and analysed during the current study are available from the corresponding author on reasonable request.

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REFERENCES

1 Francino MP. Antibiotics and the human gut microbiome: Dysbioses and accumulation of resistances. Front. Microbiol. 2016;**6**:1543. doi:10.3389/fmicb.2015.01543

2 Young VB, Schmidt TM. Antibiotic-associated diarrhea accompanied by large-scale alterations in the composition of the fecal microbiota. *J Clin Microbiol* 2004;**42**:1203–6. doi:10.1128/JCM.42.3.1203-1206.2004

3 Elseviers MM, Van Camp Y, Nayaert S, *et al.* Prevalence and management of antibiotic associated diarrhea in general hospitals. *BMC Infect Dis* 2015;**15**:129. doi:10.1186/s12879-015-0869-0

4 Bishara J, Peled N, Pitlik S, *et al.* Mortality of patients with antibiotic-associated diarrhoea: the impact of Clostridium difficile. *J Hosp Infect* 2008;**68**:308–14. doi:10.1016/J.JHIN.2008.01.033

5 Kamdeu Fansi AA, Guertin JR, LeLorier J. Savings from the use of a probiotic formula in the prophylaxis of antibiotic-associated diarrhea. *J Med Econ* 2012;**15**:53–60. doi:10.3111/13696998.2011.629015

6 Lenoir-Wijnkoop I, Nuijten MJC, Craig J, *et al.* Nutrition economic evaluation of a probiotic in the prevention of antibiotic-associated diarrhea. *Front Pharmacol* 2014;**5 FEB**. doi:10.3389/fphar.2014.00013

7 McFarland L V. Antibiotic-associated diarrhea: epidemiology, trends and treatment. *Future Microbiol* 2008;**3**:563–78. doi:10.2217/17460913.3.5.563

8 Hill C, Guarner F, Reid G, *et al.* Expert consensus document: The international scientific association for probiotics and prebiotics consensus statement on the scope and appropriate use of the term probiotic. *Nat Rev Gastroenterol Hepatol* 2014;**11**:506–14. doi:10.1038/nrgastro.2014.66

9 Naaber P, Mikelsaar M. Interactions between Lactobacilli and antibiotic-associated diarrhea. *Adv Appl Microbiol* 2004;**54**:231–60. doi:10.1016/S0065-2164(04)54009-1

10 Elmer GW, McFarland L V. Suppression by Saccharomyces boulardii of toxigenic Clostridium difficile overgrowth after vancomycin treatment in hamsters. *Antimicrob Agents Chemother* 1987;**31**:129–31. doi:10.1128/AAC.31.1.129

11 Pothoulakis C, Kelly CP, Joshi MA, *et al.* Saccharomyces boulardii inhibits Clostridium difficile toxin A binding and enterotoxicity in rat ileum. *Gastroenterology* 1993;**104**:1108–15.http://www.ncbi.nlm.nih.gov/pubmed/8462799 (accessed 7 Mar 2018).

12 Castagliuolo I, Riegler MF, Valenick L, *et al.* Saccharomyces boulardii protease inhibits the effects of Clostridium difficile toxins A and B in human colonic mucosa. *Infect Immun* 1999;**67**:302–7.http://www.ncbi.nlm.nih.gov/pubmed/9864230 (accessed 7 Mar 2018).

13 Link-Amster H, Rochat F, Saudan KY, *et al.* Modulation of a specific humoral immune response and changes in intestinal flora mediated through fermented milk intake. *FEMS Immunol Med Microbiol* 1994;**10**:55–63.http://www.ncbi.nlm.nih.gov/pubmed/7874079 (accessed 8 Mar 2018).

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14 Qamar A, Aboudola S, Warny M, *et al.* Saccharomyces boulardii Stimulates Intestinal Immunoglobulin A Immune Response to Clostridium difficile Toxin A in Mice. *Infect Immun* 2001;**69**:2762–5. doi:10.1128/IAI.69.4.2762-2765.2001

15 Yasui H, Shida K, Matsuzaki T, *et al.* Immunomodulatory function of lactic acid bacteria. *Antonie Van Leeuwenhoek* 1999;**76**:383–9.http://www.ncbi.nlm.nih.gov/pubmed/10532394 (accessed 8 Mar 2018).

16 Gill HS, Rutherfurd KJ, Cross ML. Dietary probiotic supplementation enhances natural killer cell activity in the elderly: an investigation of age-related immunological changes. *J Clin Immunol* 2001;**21**:264–71.http://www.ncbi.nlm.nih.gov/pubmed/11506196 (accessed 8 Mar 2018).

17 Wilcox MH, Cunniffe JG, Trundle C, *et al.* Financial burden of hospital-acquired Clostridium difficile infection. *J Hosp Infect* 1996;**34**:23–30.http://www.ncbi.nlm.nih.gov/pubmed/8880547 (accessed 8 Feb 2018).

18 Walk ST, Young VB. Emerging Insights into Antibiotic-Associated Diarrhea and Clostridium difficile Infection through the Lens of Microbial Ecology. *Interdiscip Perspect Infect Dis* 2008;**2008**:125081. doi:10.1155/2008/125081

19 Independent Hospital Pricing Authority. National Hospital Cost Data Collection Report. 2019;**21**.https://www.ihpa.gov.au/sites/default/files/publications/national_hospital_cost_data_collection_australian_public_hospitals_cost_report_round_21_2016-17.pdf

20 Goldenberg JZ, Yap C, Lytvyn L, *et al.* Probiotics for the prevention of Clostridium difficile-associated diarrhea in adults and children. *Cochrane database Syst Rev* 2017;**12**:CD006095. doi:https://dx.doi.org/10.1002/14651858.CD006095.pub4

21 Hurley BW, Nguyen CC. The Spectrum of Pseudomembranous Enterocolitis and Antibiotic-Associated Diarrhea. *Arch Intern Med* 2002;**162**:2177–84.https://pdfs.semanticscholar.org/cd35/9b31def5f11b39fd18f0657d24d146275ac9.pdf (accessed 1 Mar 2018).

22 Horosheva T V., Sorokulova I, Vodyanoy V. Efficacy of Bacillus probiotics in prevention of antibiotic-associated diarrhoea: a randomized, double-blind, placebo-controlled clinical trial. *JMM Case Reports* 2014;**1**:1–6. doi:10.1099/jmmcr.0.004036

23 Moher D, Liberati A, Tetzlaff J, *et al.* Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement (Reprinted from Annals of Internal Medicine). *PLoS Med* 2009;**6**:1–6. doi:10.1371/journal.pmed.1000097

24 Higgins J, Altman D. Chapter 8: Assessing risk of bias in included studies Table 8.5 a. In: *Cochrane Handbook for Systematic Reviews of ….* The Cochrane Collaboration, Availa 2008. 1–50. doi:10.1001/jama.2013.109501.Conflict

25 Guyatt GH, Oxman AD, Vist GE, *et al.* GRADE: An emerging consensus on rating quality of evidence and strength of recommendations. *BMJ* 2008;**336**:924–6. doi:10.1136/bmj.39489.470347.ad

26 Allen SJ, Wareham K, Wang D, *et al.* Lactobacilli and bifidobacteria in the prevention of antibiotic-associated diarrhoea and Clostridium difficile diarrhoea in older inpatients

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(PLACIDE): a randomised, double-blind, placebo-controlled, multicentre trial. *Lancet* 2013;**382**:1249–57. doi:10.1016/S0140-6736(13)61218-0

27 Beausoleil M, Fortier N, Guénette S, *et al.* Effect of a fermented milk combining Lactobacillus acidophilus CL1285 and Lactobacillus casei in the prevention of antibiotic-associated diarrhea: A randomized, double-blind, placebo-controlled trial. *Can J Gastroenterol* 2007;**21**:732–6.http://www.ncbi.nlm.nih.gov/pubmed/18026577 (accessed 6 Dec 2017).

28 Beniwal RS, Arena VC, Thomas L, *et al.* A randomized trial of yogurt for prevention of antibiotic-associated diarrhea. *Dig Dis Sci* 2003;**48**:2077–82. doi:10.1023/A:1026155328638

29 Bravo MV, Bunout D, Leiva L, *et al.* Efecto del probiótico Saccharomyces boulardii en la prevención de la diarrea asociada con antibióticos en adultos ambulatorios en tratamiento con amoxicilina. *Rev Med Chil* 2008;**136**:981–8. doi:10.4067/s0034-98872008000800004

30 Can M, Beşirbellioglu BA, Avci IY, *et al.* Prophylactic Saccharomyces boulardii in the prevention of antibiotic-associated diarrhea: a prospective study. *Med Sci Monit* 2006;**12**:PI19-I22. doi:6635 [pii]

31 Chatterjee S, Kar P, Das T, *et al.* Randomised placebo-controlled double blind multicentric trial on efficacy and safety of Lactobacillus acidophilus LA-5 and Bifidobacterium BB-12 for prevention of antibiotic-associated diarrhoea. *J Assoc Physicians India* 2013;**61**:708–12.http://www.ncbi.nlm.nih.gov/pubmed/24772726 (accessed 6 Dec 2017).

32 Cimperman L, Bayless G, Best K, *et al.* A Randomized, Double-blind, Placebo-controlled Pilot Study of Lactobacillus reuteri ATCC 55730 for the Prevention of Antibiotic-associated Diarrhea in Hospitalized Adults. *J Clin Gastroenterol* 2011;**45**:785–9. doi:10.1097/MCG.0b013e3182166a42

33 Cindoruk M, Erkan G, Karakan T, *et al.* Efficacy and safety of Saccharomyces boulardii in the 14-day triple anti-Helicobacter pylori therapy: A prospective randomized placebo-controlled double-blind study. *Helicobacter* 2007;**12**:309–16. doi:10.1111/j.1523-5378.2007.00516.x

34 de Vrese M, Kristen H, Rautenberg P, *et al.* Probiotic lactobacilli and bifidobacteria in a fermented milk product with added fruit preparation reduce antibiotic associated diarrhea and Helicobacter pylori activity. *J Dairy Res* 2011;**78**:396–403. doi:10.1017/S002202991100063X

35 Duman DG, Bor S, Ozütemiz O, *et al.* Efficacy and safety of Saccharomyces boulardii in prevention of antibiotic-associated diarrhoea due to Helicobacterpylori eradication. *Eur J Gastroenterol Hepatol* 2005;**17**:1357–61.http://www.ncbi.nlm.nih.gov/pubmed/16292090 (accessed 6 Dec 2017).

36 Ehrhardt S, Guo N, Hinz R, *et al.* Saccharomyces boulardii to prevent antibiotic-associated diarrhea: A randomized, double-masked, placebo-controlled trial. *Open Forum Infect Dis* 2016;**3**:1–7. doi:10.1093/ofid/ofw011

37 Evans M, Salewski RP, Christman MC, *et al.* Effectiveness of Lactobacillus helveticus and Lactobacillus rhamnosus for the management of antibiotic-associated diarrhoea in healthy adults: a randomised, double-blind, placebo-controlled trial. *Br J Nutr* 2016;**116**:94–103. doi:10.1017/S0007114516001665

38 Gao XW, Mubasher M, Fang CY, *et al.* Dose-response efficacy of a proprietary probiotic

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formula of Lactobacillus acidophilus CL1285 and Lactobacillus casei LBC80R for antibiotic-associated diarrhea and Clostridium difficile-associated diarrhea prophylaxis in adult patients. *Am J Gastroenterol* 2010;**105**:1636–41. doi:10.1038/ajg.2010.11

39 Gotz V, Romankiewicz JA, Moss J, *et al.* Prophylaxis against ampicillin-associated diarrhea with a lactobacillus preparation. *Am J Hosp Pharm* 1979;**36**:754–7.http://ovidsp.ovid.com/ovidweb.cgi?T=JS&PAGE=reference&D=med1&NEWS=N&AN=111546

40 Helps A, Bell E, Mactier R. Prospective randomized double-blind study of efficacy of probiotic milk drink in reducing the incidence of antibiotic-associated diarrhoea and clostridium difficile Diarrhoea. *Int J Probiotics Prebiotics* 2015;**11**:145–52.

41 Hickson M, D'Souza AL, Muthu N, *et al.* Use of probiotic Lactobacillus preparation to prevent diarrhoea associated with antibiotics: randomised double blind placebo controlled trial. *BMJ* 2007;**335**:80. doi:10.1136/bmj.39231.599815.55

42 Iamharit J, Harnsomburana P. Prophylactic Use of Probiotic (Infloran ®) to Prevent Antibiotic Associated Diarrhea in Medical Ward Patients at Rajavithi Hospital. *Thai J Gastroenterol* 2010;**11**:155–60.

43 Imase K, Takahashi M, Tanaka A, *et al.* Efficacy of Clostridium butyricum preparation concomitantly with Helicobacter pylori eradication therapy in relation to changes in the intestinal microbiota. *Microbiol Immunol* 2008;**52**:156–61. doi:https://dx.doi.org/10.1111/j.1348-0421.2008.00026.x

44 Khanal A, Mishra A, Shrestha D, *et al.* Efficacy Of A Mixture Of Probiotic Containing saccharomyces boulardii and lactobacillus in Preventing Antibiotic Associated Diarrhea In Patients Taking Triple Therapy For H Pylori Eradication. *Post-Graduate Med J NAMS* 2015;**15**.https://pmjn.org.np/index.php/pmjn/article/view/268 (accessed 10 Jul 2019).

45 Koning CJM, Jonkers DMAE, Stobberingh EE, *et al.* The effect of a multispecies probiotic on the intestinal microbiota and bowel movements in healthy volunteers taking the antibiotic amoxycillin. *Am J Gastroenterol* 2008;**103**:178–89. doi:10.1111/j.1572-0241.2007.01547.x

46 Lewis SJ, Potts LF, Barry RE. The lack of therapeutic effect of Saccharomyces boulardii in the prevention of antibiotic-related diarrhoea in elderly patients. *J Infect* 1998;**36**:171–4. doi:10.1016/S0163-4453(98)80008-X

47 Lönnermark E, Friman V, Lappas G, *et al.* Intake of Lactobacillus plantarum Reduces Certain Gastrointestinal Symptoms During Treatment With Antibiotics. *J Clin Gastroenterol* 2010;**44**:106–12. doi:10.1097/MCG.0b013e3181b2683f

48 McFarland L V., Surawicz CM, Greenberg RN, *et al.* Prevention of β-Lactam-Associated Diarrhea by Saccharomyces boulardii Compared with Placebo. *Am J Gastroenterol* 1995;**90**:439–48. doi:10.1111/j.1572-0241.1995.tb08241.x

49 Ouwehand AC, DongLian C, Weijian X, *et al.* Probiotics reduce symptoms of antibiotic use in a hospital setting: a randomized dose response study. *Vaccine* 2014;**32**:458–63. doi:10.1016/j.vaccine.2013.11.053

50 Pirker A, Stockenhuber A, Remely M, *et al.* Effects of antibiotic therapy on the gastrointestinal

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microbiota and the influence of *Lactobacillus casei*. *Food Agric Immunol* 2013;**24**:315–30. doi:10.1080/09540105.2012.689816

51 Plomer M, Perez M, Greifenberg DM. Effect of Bacillus clausii Capsules in Reducing Adverse Effects Associated with Helicobacter pylori Eradication Therapy: A Randomized, Double-Blind, Controlled Trial. *Infect Dis Ther* 2020;**9**:867–78. doi:10.1007/s40121-020-00333-2

52 Plummer S, Weaver MA, Harris JC, *et al.* Clostridium difficile pilot study: effects of probiotic supplementation on the incidence of C-difficile diarrhoea. *Int Microbiol* 2004;**7**:59–62.file:///Users/clarelouisagoodman/Downloads/09 Plummer.pdf

53 Pozzoni P, Riva A, Bellatorre AG, *et al.* Saccharomyces boulardii for the prevention of antibiotic-associated diarrhea in adult hospitalized patients: a single-center, randomized, double-blind, placebo-controlled trial. *Am J Gastroenterol* 2012;**107**:922–31. doi:10.1038/ajg.2012.56

54 Rajkumar C, Wilks M, Islam J, *et al.* Do probiotics prevent antibiotic-associated diarrhoea? Results of a multicentre randomized placebo-controlled trial. *J Hosp Infect* 2020;**105**:280–8. doi:10.1016/j.jhin.2020.01.018

55 Safdar N, Barigala R, Said A, *et al.* Feasibility and tolerability of probiotics for prevention of antibiotic-associated diarrhoea in hospitalized US military veterans. *J Clin Pharm Ther* 2008;**33**:663–8. doi:10.1111/j.1365-2710.2008.00980.x

56 Sampalis J, Psaradellis E, Rampakakis E. Efficacy of BIO K+ CL1285 in the reduction of antibiotic-associated diarrhea - a placebo controlled double-blind randomized, multi-center study. *Arch Med Sci* 2010;**6**:56–64. doi:10.5114/aoms.2010.13508

57 Selinger CPP, Bell A, Cairns A, *et al.* Probiotic VSL#3 prevents antibiotic-associated diarrhoea in a double-blind, randomized, placebo-controlled clinical trial. *J Hosp Infect* 2013;**84**:159–65. doi:10.1016/j.jhin.2013.02.019

58 Shimbo I, Yamaguchi T, Odaka T, *et al.* Effect of Clostridium butyricum on fecal flora in Helicobacter pylori eradication therapy. *World J Gastroenterol* 2005;**11**:7520–4. doi:10.3748/wjg.v11.i47.7520

59 Song HJ, Kim J-Y, Jung S-A, *et al.* Effect of probiotic Lactobacillus (Lacidofil® cap) for the prevention of antibiotic-associated diarrhea: a prospective, randomized, double-blind, multicenter study. *J Korean Med Sci* 2010;**25**:1784–91. doi:10.3346/jkms.2010.25.12.1784

60 Surawicz CM, Elmer GW, Speelman P, *et al.* Prevention of antibiotic-associated diarrhea by Saccharomyces boulardii: a prospective study. *Gastroenterology* 1989;**96**:981–8.http://www.ncbi.nlm.nih.gov/pubmed/2494098 (accessed 6 Dec 2017).

61 Thomas MR, Litin SC, Osmon DR, *et al.* Lack of effect of Lactobacillus GG on antibiotic-associated diarrhea: a randomized, placebo-controlled trial. *Mayo Clin Proc* 2001;**76**:883–9. doi:10.4065/76.9.883

62 Velasco M, Requena T, Delgado-Iribarren A, *et al.* Probiotic Yogurt for the Prevention of Antibiotic-associated Diarrhea in Adults: A Randomized Double-blind Placebo-controlled Trial. *J Clin Gastroenterol* 2019;**53**:717–23. doi:10.1097/MCG.0000000000001131

63 Wenus C, Goll R, Loken EB, *et al.* Prevention of antibiotic-associated diarrhoea by a

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fermented probiotic milk drink. *Eur J Clin Nutr* 2008;**62**:299–301. doi:10.1038/sj.ejcn.1602718

64 Wong S, Jamous A, O'Driscoll J, *et al.* A Lactobacillus casei Shirota probiotic drink reduces antibiotic-associated diarrhoea in patients with spinal cord injuries: a randomised controlled trial. *Br J Nutr* 2013;**111**:672–8. doi:S0007114513002973 [pii]\r10.1017/S0007114513002973

65 Wright K, Wright H, Murray M. Probiotic treatment for the prevention of antibiotic-associated diarrhoea in geriatric patients: a multicentre randomised controlled pilot study. *Australas J Ageing* 2015;**34**:38–42. doi:10.1111/ajag.12116

66 Wu J, Gan T, Zhang Y, *et al.* The prophylactic effects of BIFICO on the antibiotic-induced gut dysbiosis and gut microbiota. *Gut Pathog* 2020;**12**:1–11. doi:10.1186/s13099-020-00379-0

67 Higgins J, Thomas J, Chandler J, *et al. Cochrane Handbook for Systematic Reviews of Interventions*. 2019. doi:10.1002/9781119536604

68 Dietrich CG, Kottmann T, Alavi M. Commercially available probiotic drinks containing Lactobacillus casei DN-114001 reduce antibiotic-associated diarrhea. *World J Gastroenterol* 2014;**20**:15837–44. doi:10.3748/wjg.v20.i42.15837

69 Price DD, Finniss DG, Benedetti F. A Comprehensive Review of the Placebo Effect: Recent Advances and Current Thought. *Annu Rev Psychol* 2008;**59**:565–90. doi:10.1146/annurev.psych.59.113006.095941

70 Hempel S, Newberry SJ, Maher AR, *et al.* Probiotics for the prevention and treatment of antibiotic-associated diarrhea: a systematic review and meta-analysis. *JAMA* 2012;**307**:1959–69. doi:10.1001/jama.2012.3507

71 Jafarnejad S, Shab-Bidar S, Speakman JR, *et al.* Probiotics Reduce the Risk of Antibiotic-Associated Diarrhea in Adults (18–64 Years) but Not the Elderly (>65 Years). *Nutr Clin Pract* 2016;**31**:502–13. doi:10.1177/0884533616639399

72 Su GL, Ko CW, Bercik P, *et al.* AGA Clinical Practice Guidelines on the Role of Probiotics in the Management of Gastrointestinal Disorders. *Gastroenterology* 2020;**159**:697–705. doi:10.1053/j.gastro.2020.05.059

73 Xie C, Li J, Wang K, *et al.* Probiotics for the prevention of antibiotic- associated diarrhoea in older patients: A systematic review. *Travel Med Infect Dis* 2015;**13**:128–34. doi:10.1016/j.tmaid.2015.03.001

Figure 1: PRISMA flow diagram depicting the phases of this systematic review. A total of 352 articles were identified in the initial search, 250 remained after removal of duplicates, of which 60 were eligible for full-text review, and 42 of those articles were eligible for inclusion in the study [23].

Figure 2: Risk of bias summary: review authors' judgements about each risk of bias item for each included study

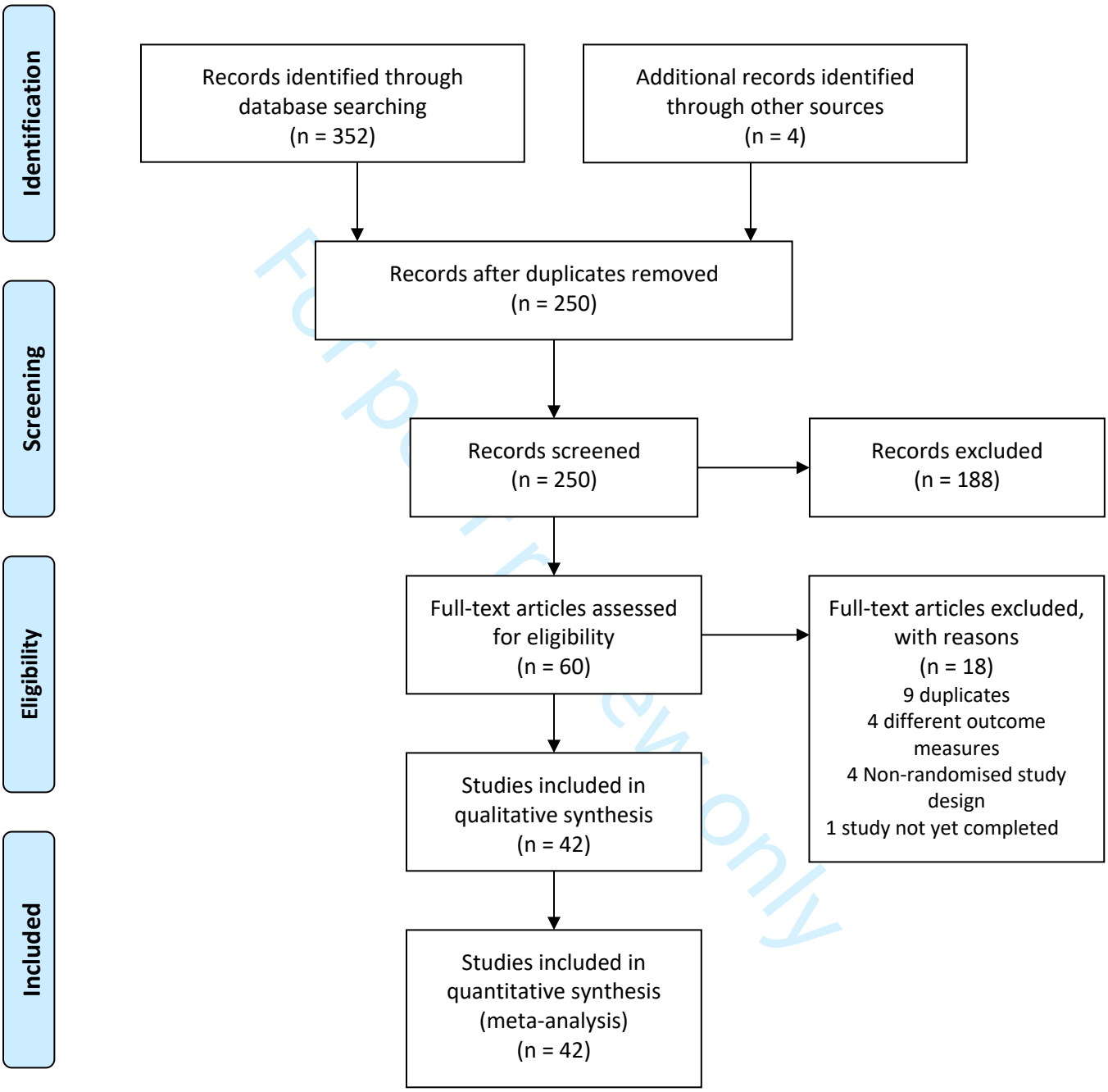
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Figure 3: Forest plot of comparison: 1 Probiotic vs any control, Outcome: 1.1 Incidence AAD: complete case.

Figure 4: Funnel plot of comparison: 1 Probiotic vs any control, Outcome: 1.1 Incidence AAD: complete case.

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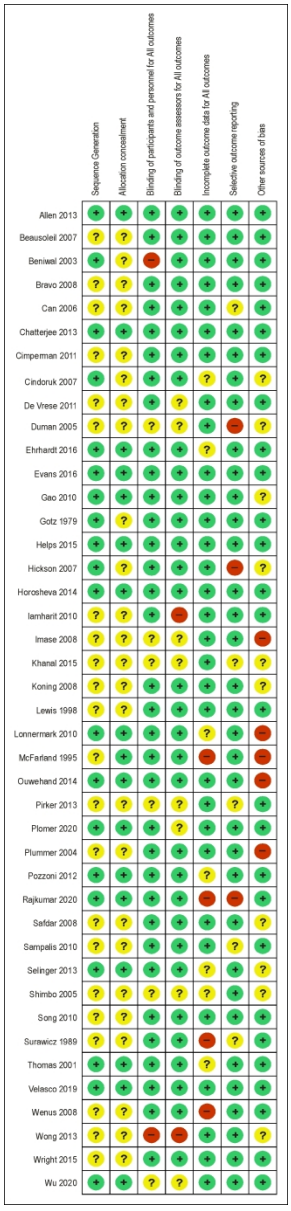


Figure 2: Risk of bias summary: review authors' judgements about each risk of bias item for each included study.

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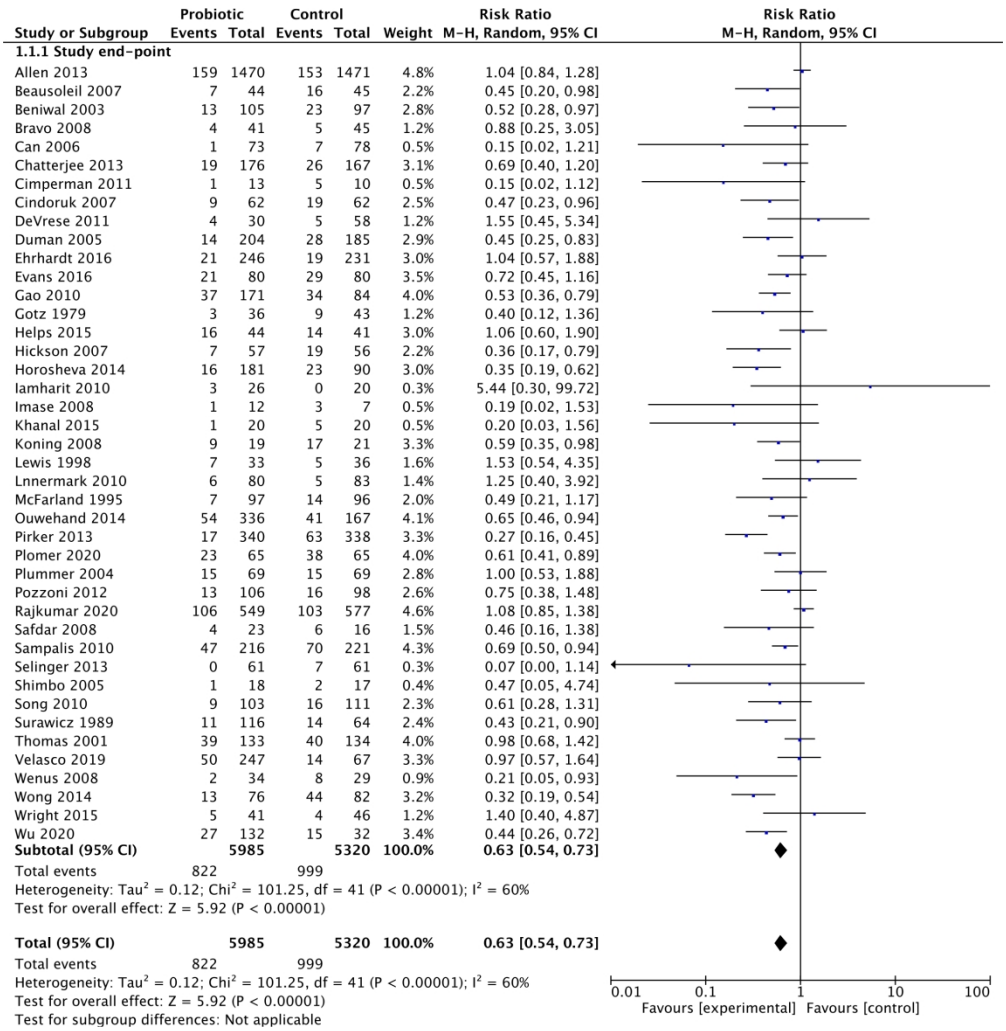


Figure 3: Forest plot of comparison: 1 Probiotic vs any control, Outcome: 1.1 Incidence AAD: complete case.

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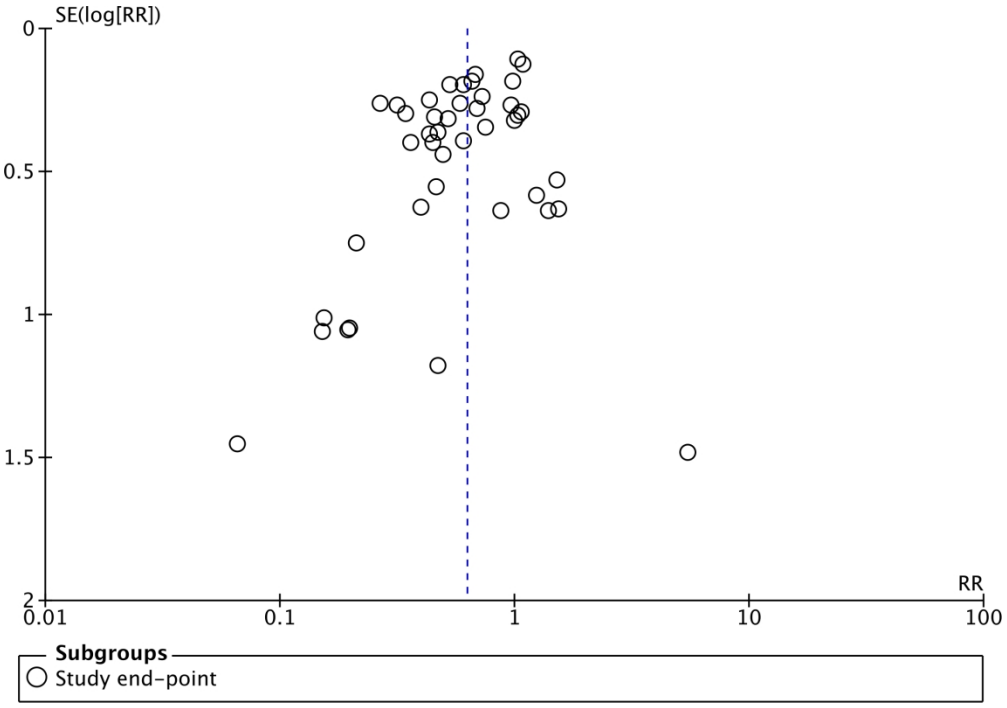


Figure 4: Funnel plot of comparison: 1 Probiotic vs any control, Outcome: 1.1 Incidence AAD: complete case.

191x133mm (300 x 300 DPI)

Supplemental Files

Supplemental file 1: Original study protocol

Review title
Probiotics for the prevention of antibiotic-associated diarrhoea: A systematic review and meta-analysis

Anticipated or actual start date
08/08/2019

Anticipated completion date
30/05/2021

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Funding sources/sponsors
NA

Conflicts of interest
None known

Collaborators
NA

Review question(s)
(1) Do probiotics reduce the incidence of antibiotic-associated diarrhoea in adults?
(2) Is there a dose-response?
(3) Is there a species-specific response?

Searches
Databases to be searched: CINAHL Plus, EMBASE, MEDLINE (Ovid), Web of Science and Google Scholar from inception to May 2021, as well as a manual search of clinical trial registries and the reference lists of relevant systematic reviews. Additional details about the search strategy can be found in the attached full search strategy.

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Condition or domain being studied

Antibiotics are one of the most commonly prescribed medications. They disturb the normal gastrointestinal microbiota, which can lead to diarrhoea. Probiotics are live microorganisms that, when administered in adequate amounts, confer a health benefit on the host. Our objective was to evaluate existing evidence for the use of probiotics in preventing antibioticassociated diarrhoea (AAD) in adults.

Participants/population

Inclusion: Studies of participants 18 years or older who were receiving any antibiotic therapy for any reason
Exclusion: Studies with participants younger than 18

Intervention(s), exposure(s)

Probiotics - any species/strain, dose or formulation

Comparator(s)/control

Placebo, alternative dose, alternative probiotic strain, or no treatment control

Outcome measure(s)

Incidence of diarrhoea as defined by authors during the antibiotic treatment or follow-up phases
Studies were excluded if they examined probiotics only for treatment, rather than prevention, of AAD

Types of study to be included

Randomised controlled trials (RCTs) only
Written in English

Context

Any setting: inpatient or outpatient

Data extraction (selection and coding)

Data will be extracted from each article by two of three of the authors (CG, GK, KL), using a standardised data extraction form following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines. This will be housed within Covidence online software (Covidence systematic review software, Veritas Health Innovation, Melbourne, Australia. Available at www.covidence.org). Data items collected to be included: number of patients allocated to each group, probiotic species, formulation, dose and regime, whether a placebo or 'nothing' control was used, and outcome data on incidence of AAD.

Risk of bias (quality) assessment

Risk of bias will assessed using the Cochrane Risk of Bias [24] tool within the Covidence online software. Assessment will be conducted by two of three authors (CG, GK, KL) and discrepancies resolved by consensus.

Strategy for data synthesis

For data analysis we will use the RevMan 5.4.1 statistical software (Review Manager (RevMan) [Computer program]. Version 5.4. The Cochrane Collaboration, 2020.) We will use either a fixed-effects or random-effects model, reported as either odds ratios or risk ratios with corresponding 95% confidence intervals (95% CI). The number needed to treat (NNT) will also calculated for the complete case analysis.

Analysis of subgroups or subsets

Our a priori clinical heterogeneity variables will include probiotic dose and probiotic species.

Type and method of review

Systematic review and meta-analysis

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Supplemental file 2: Search strategy

Electronic sources

- CINAHL Plus
- EMBASE
- MEDLINE (Ovid)
- Web of Science
- Google Scholar
- Manual search of grey literature: ○ Reference lists of relevant systematic reviews
  - ○ Clinical trial registries (https://clinicaltrials.gov/ and http://www.anzctr.org.au/)

Boolean keywords / search terms

Population
(Adults or humans or aged or elderly or young adults)

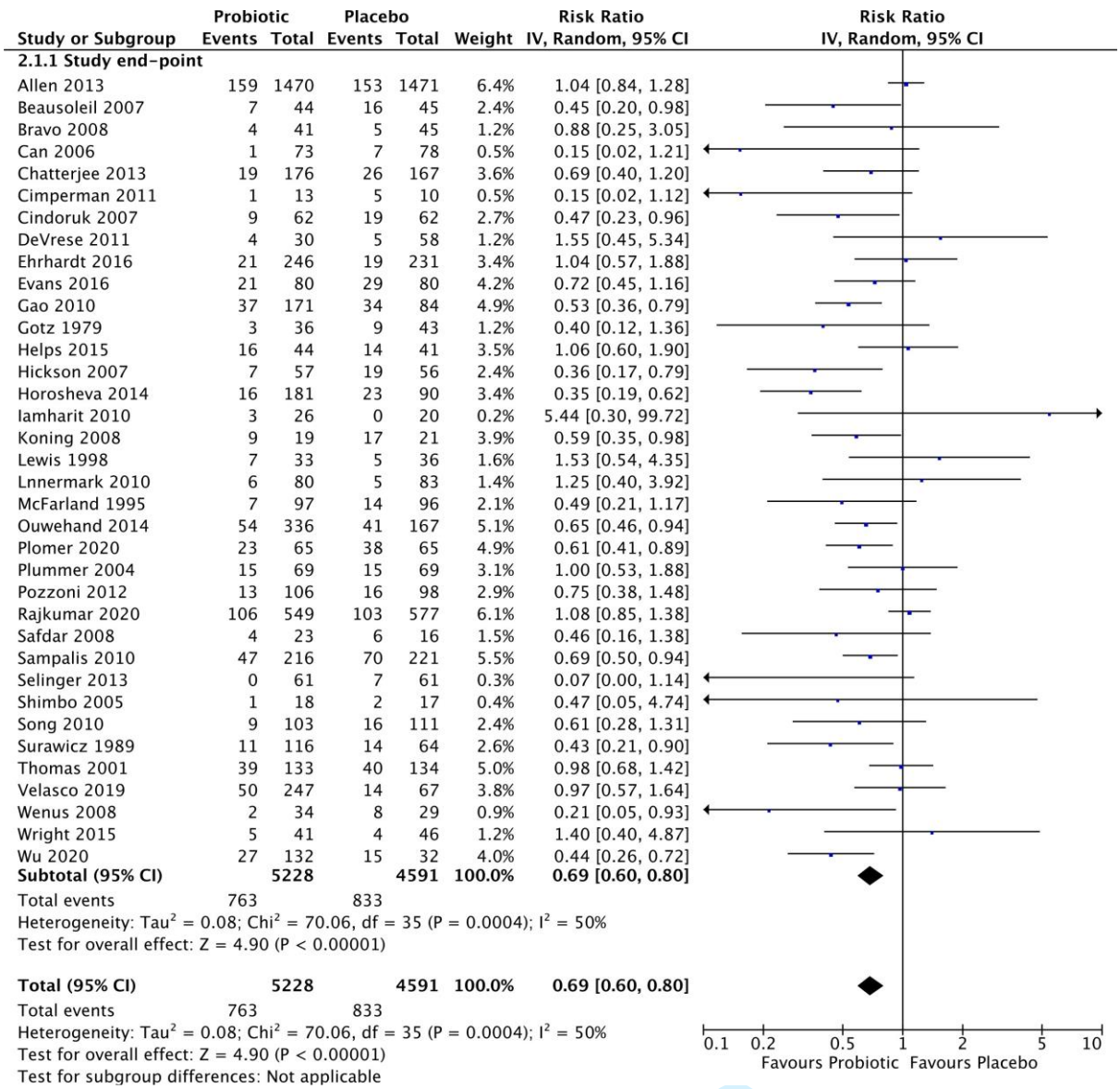
Intervention
(Probiotic or fermented milk or fermented yoghurt or lactic acid bacteria or lactobacillus or bifidobacterium or saccharomyces or lactococcus or enterococcus or streptococcus or pediococcus or leuconostoc or bacillus or escherichia or coli or rhamnosus or acidophilus or bifidum or lactis or bulgaricus or reuteri or casei or thermophilus or boulardii or coagulans or delbrueckii or animalis or thermophilus). m_titl.

Control
(Placebo or no treatment or pasteurised yoghurt or heat-treated yoghurt or control or compar*). m_titl.

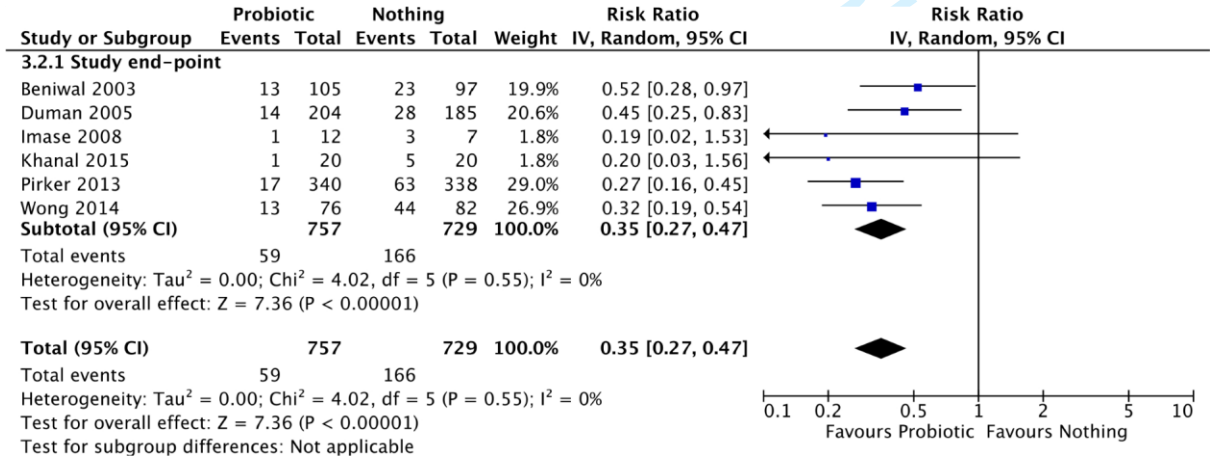
Outcome
(Incidence or prevalence or occurrence or reduction or improvement or prevention or alleviation or episode* or occasion) AND (Diarrh* or antibiotic-associated diarrh* or clostridium difficile associated diarrh* or pseudomembranous colitis or nosocomial diarrh* or infectious diarrh*). m_titl.

Supplemental file 3: Forest plots of all sub-group analyses

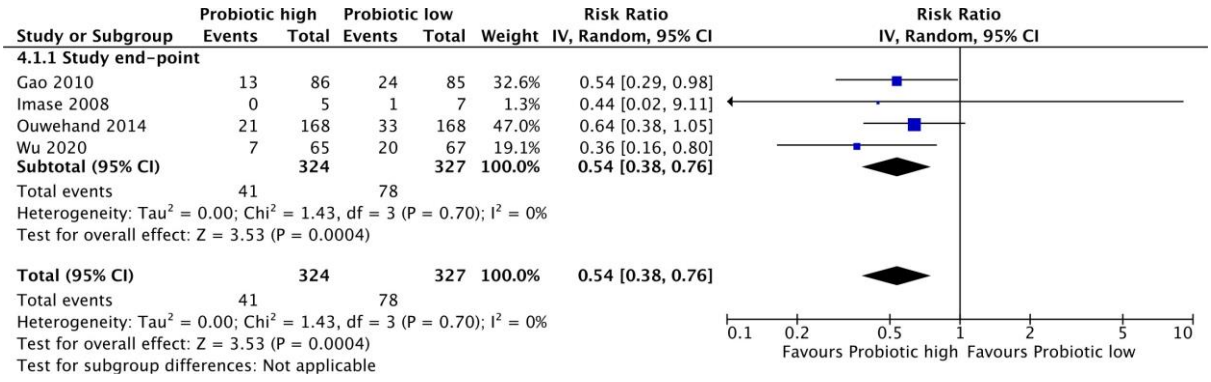
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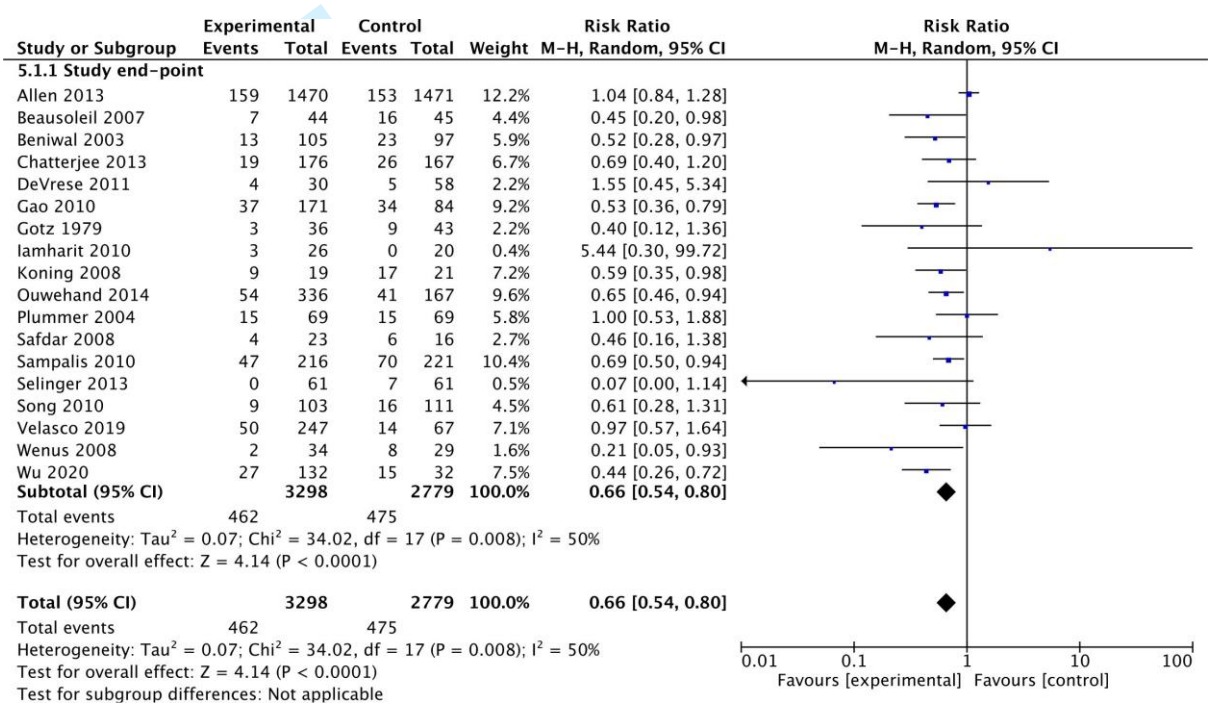
Supplemental figure 1: Subgroup analysis: Placebo effect: Probiotics vs placebo control



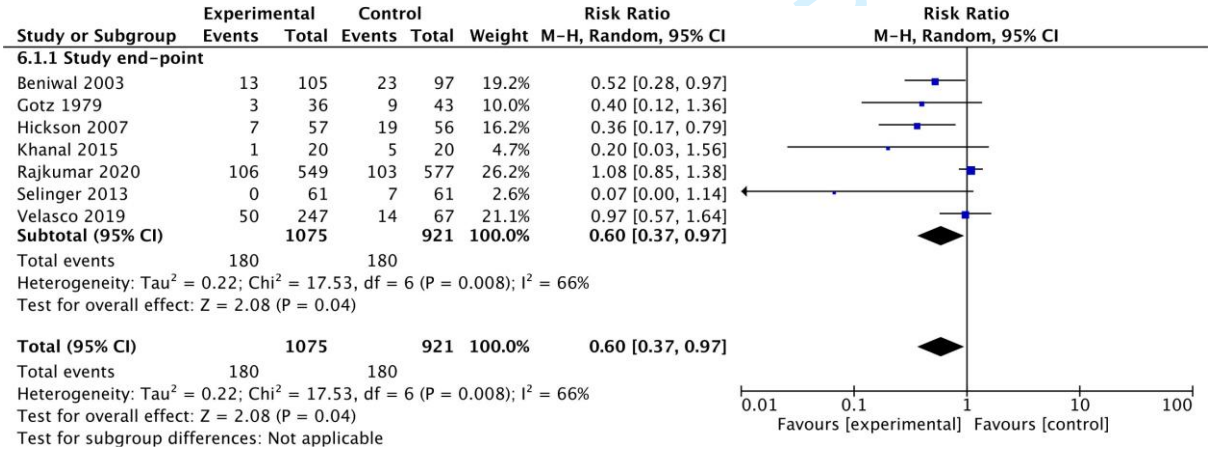
Supplemental figure 2: Subgroup analysis: Placebo effect: Probiotics vs no treatment control



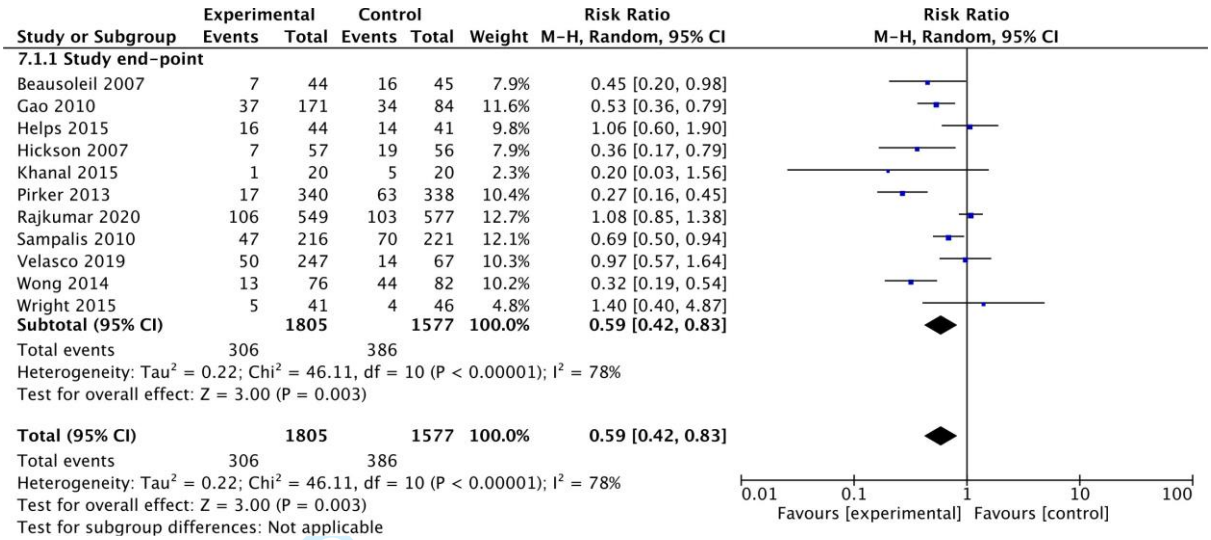
Supplemental figure 3: Subgroup analysis: dose response: High dose vs low dose



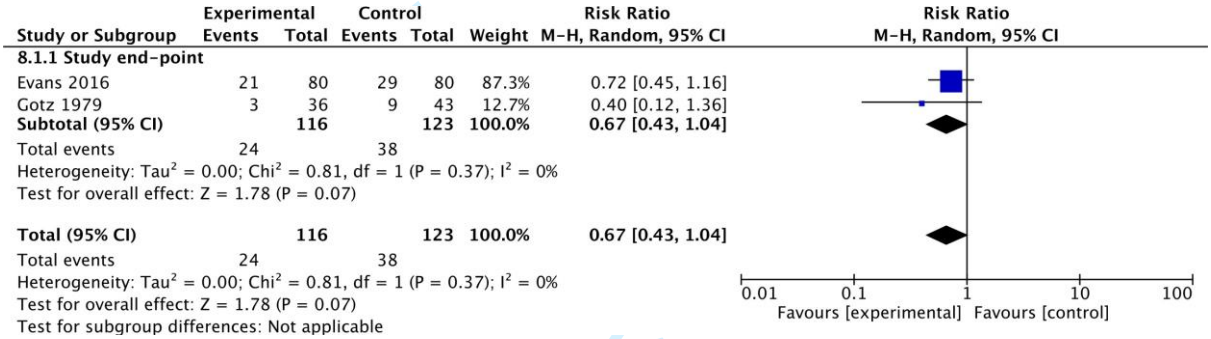
Supplemental figure 4: Subgroup analysis: species: L. acidophilus vs any control



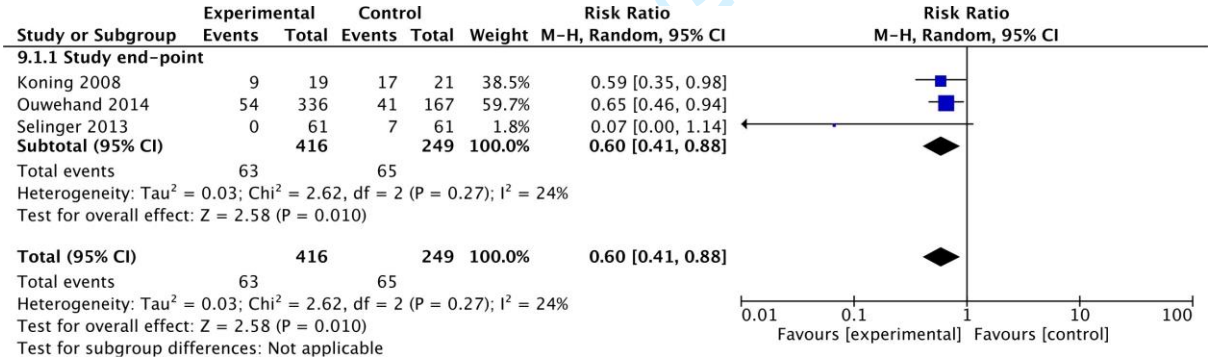
Supplemental figure 5: Subgroup analysis: species: L. bulgaricus vs any control



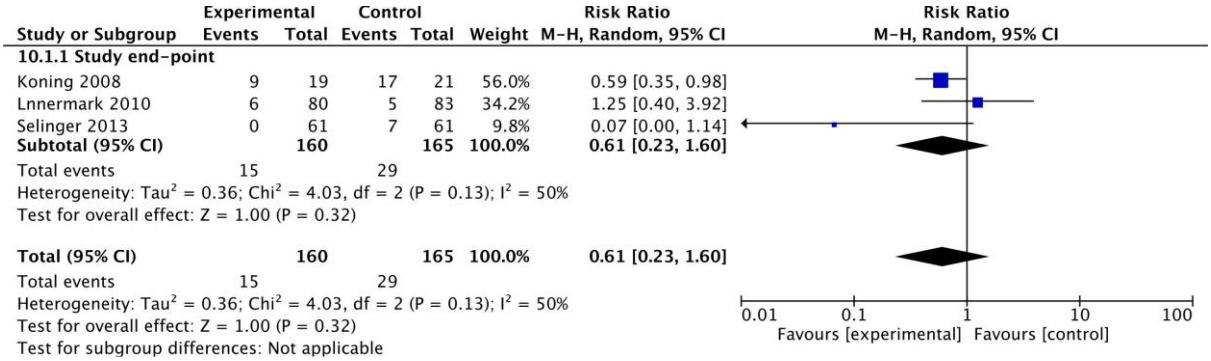
Supplemental figure 6: Subgroup analysis: species: L. casei vs any control



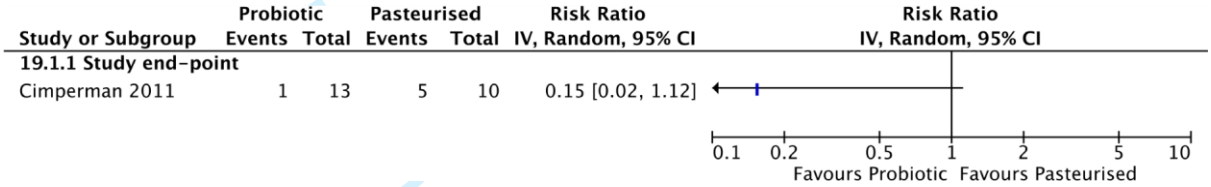
Supplemental figure 7: Subgroup analysis: species: L. helveticus vs any control



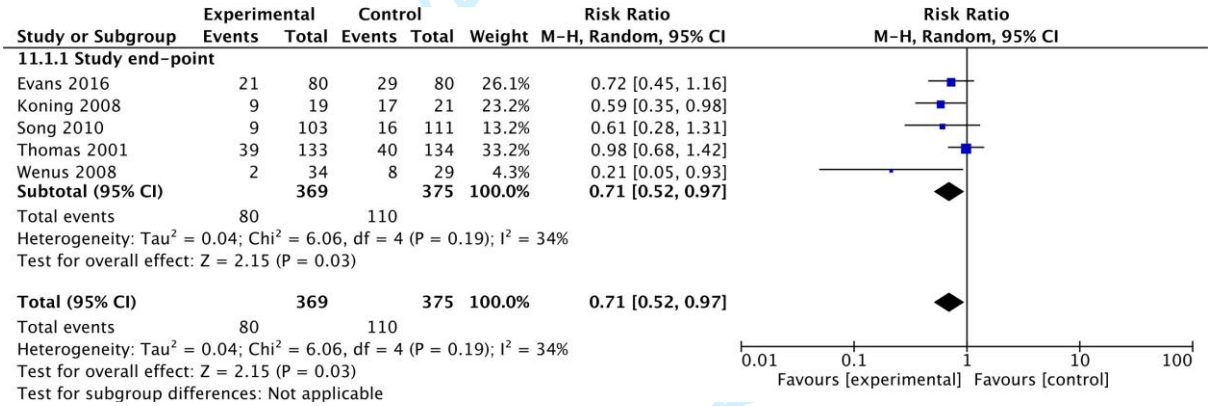
Supplemental figure 8: Subgroup analysis: species: L. paracasei vs any control



Supplemental figure 9: Subgroup analysis: species: L. plantarum vs any control

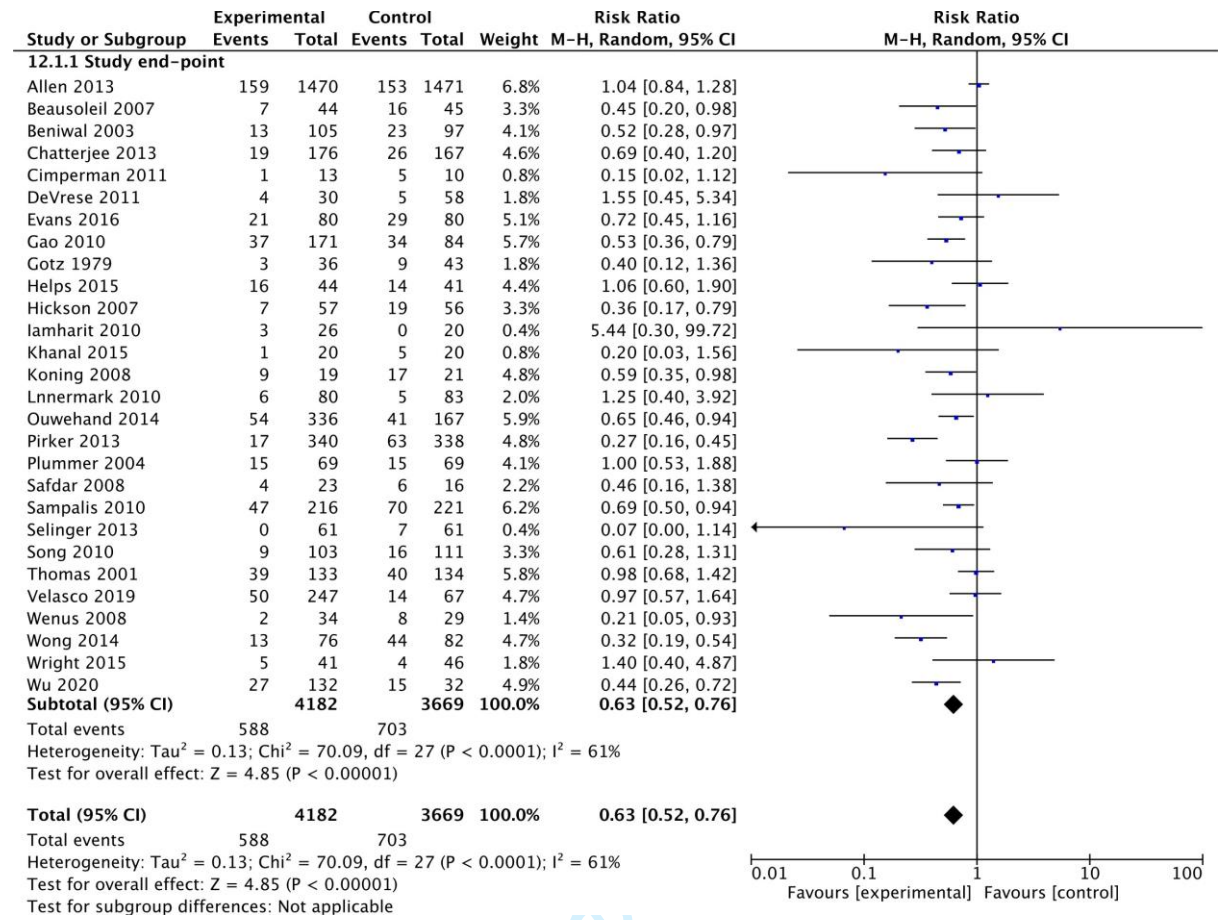


Supplemental figure 10: Subgroup analysis: species: L. reuteri vs any control

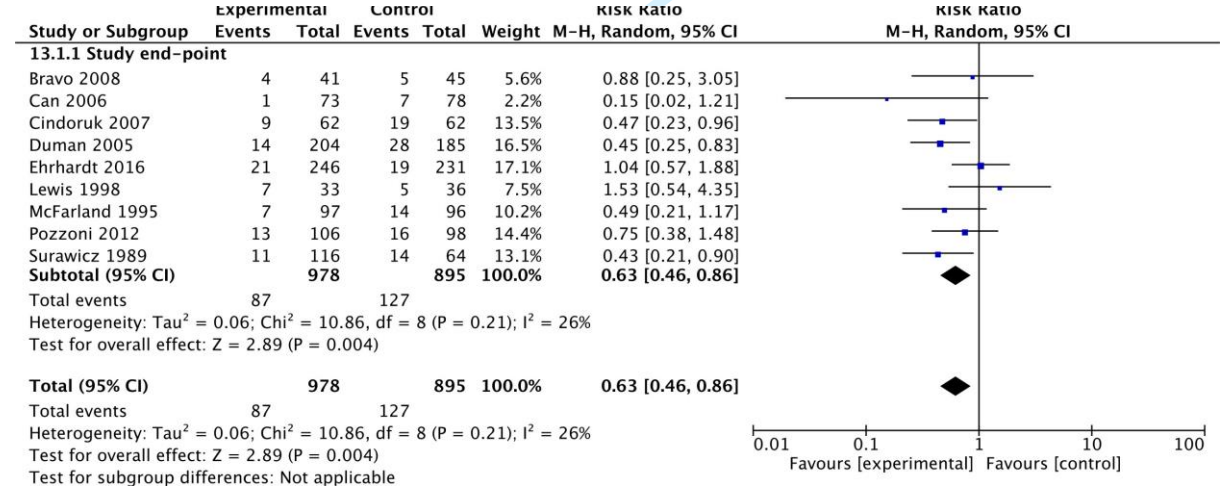


Supplemental figure 11: Subgroup analysis: species: L. rhamnosus vs any control

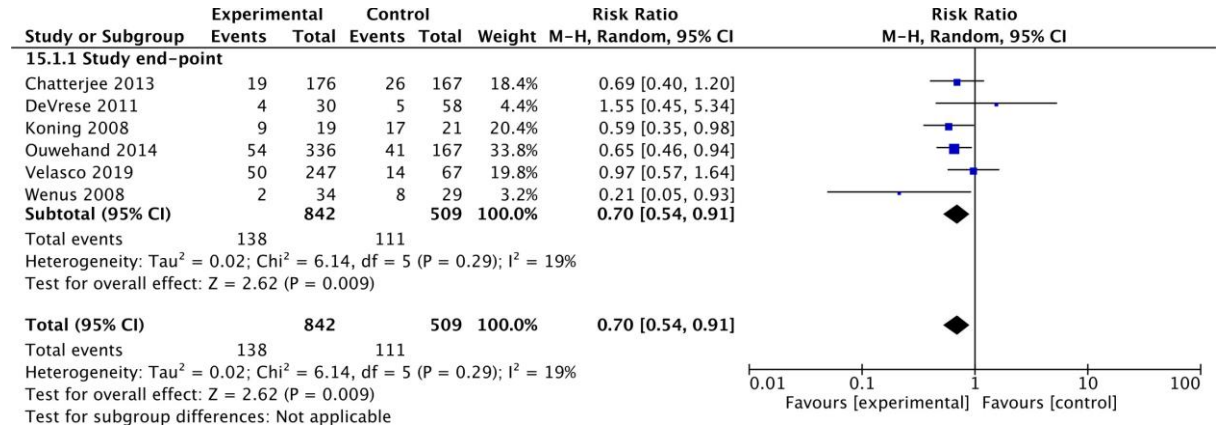
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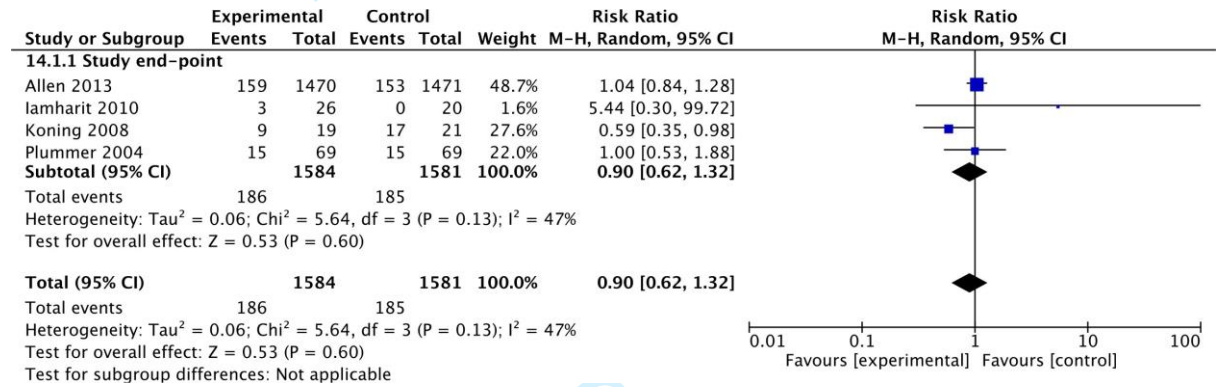
Supplemental figure 12: Subgroup analysis: species: Lactobacillus sp. vs any control



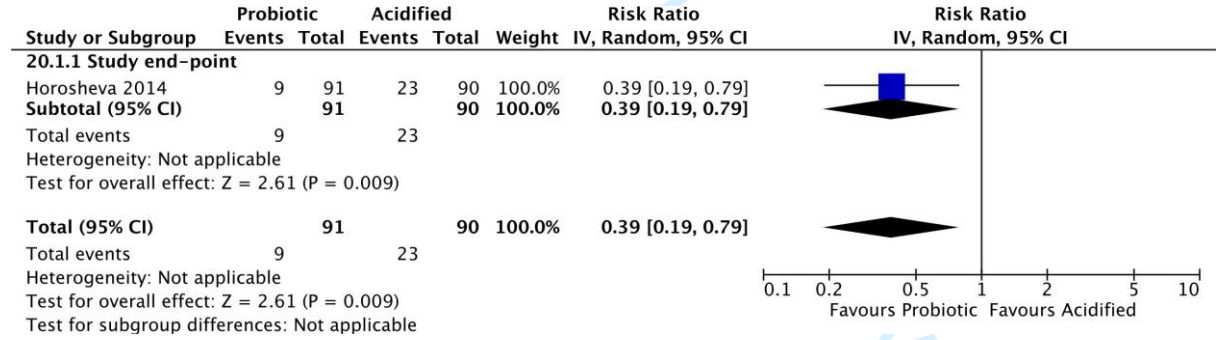
Supplemental figure 13: Subgroup analysis: species: S. boulardii vs any control



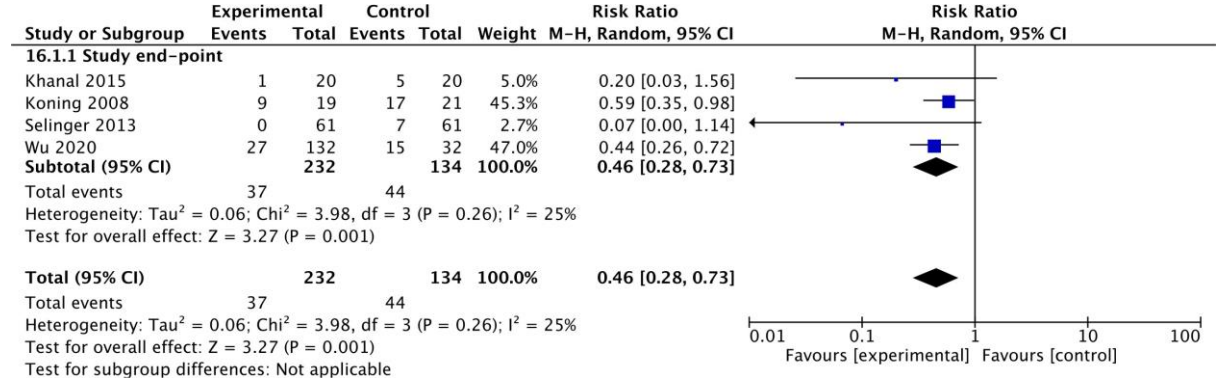
Supplemental figure 14: Subgroup analysis: species: B. animalis ssp. Lactis vs any control



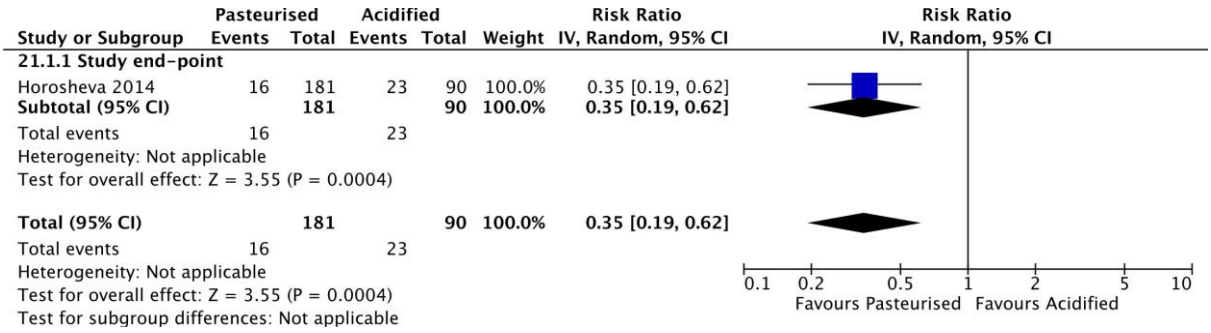
Supplemental figure 15: Subgroup analysis: species: B. bifidum vs any control



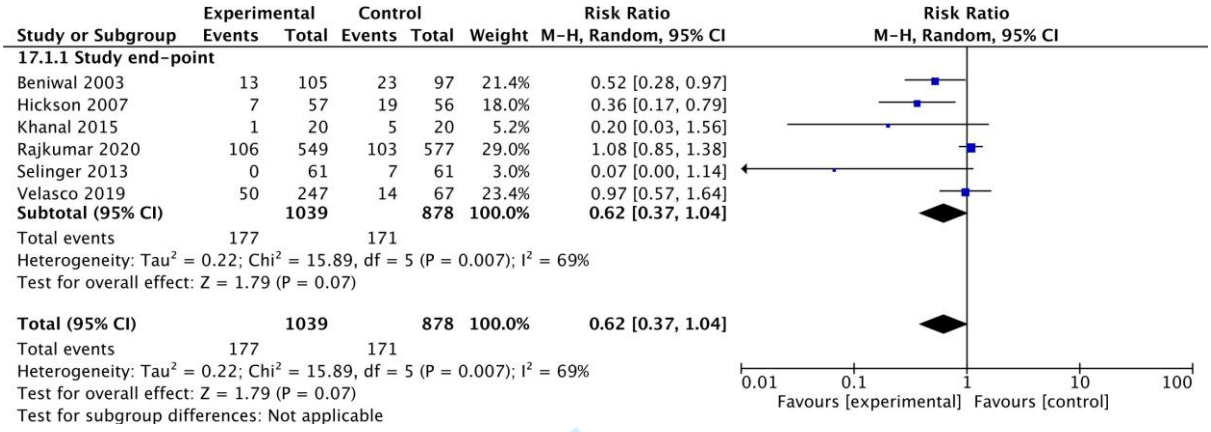
Supplemental figure 16: Subgroup analysis: species: B. licheniformis vs any control



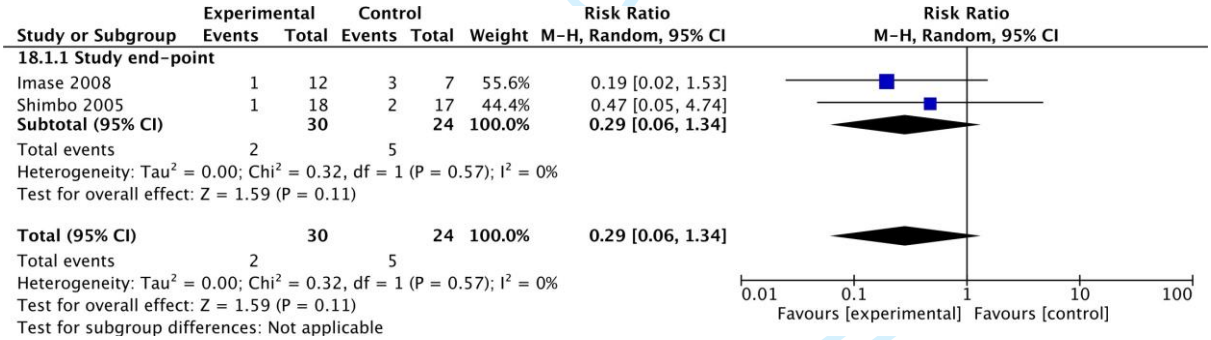
Supplemental figure 17: Subgroup analysis: species: B. longum vs any control



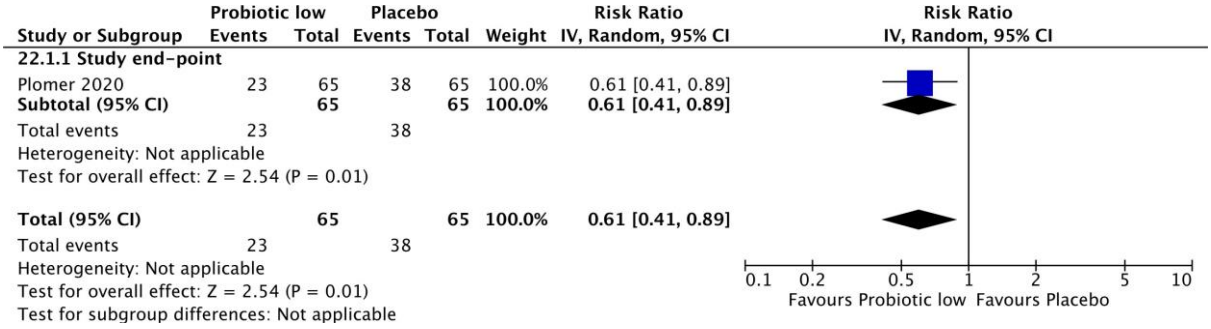
Supplemental figure 18: Subgroup analysis: species: B. subtilis vs any control



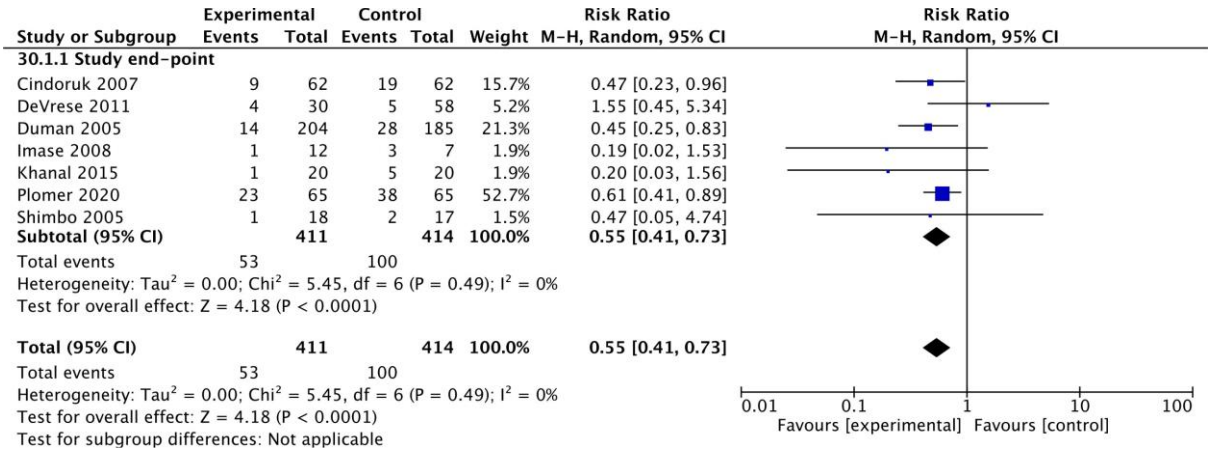
Supplemental figure 19: Subgroup analysis: species: S. thermophilus vs any control



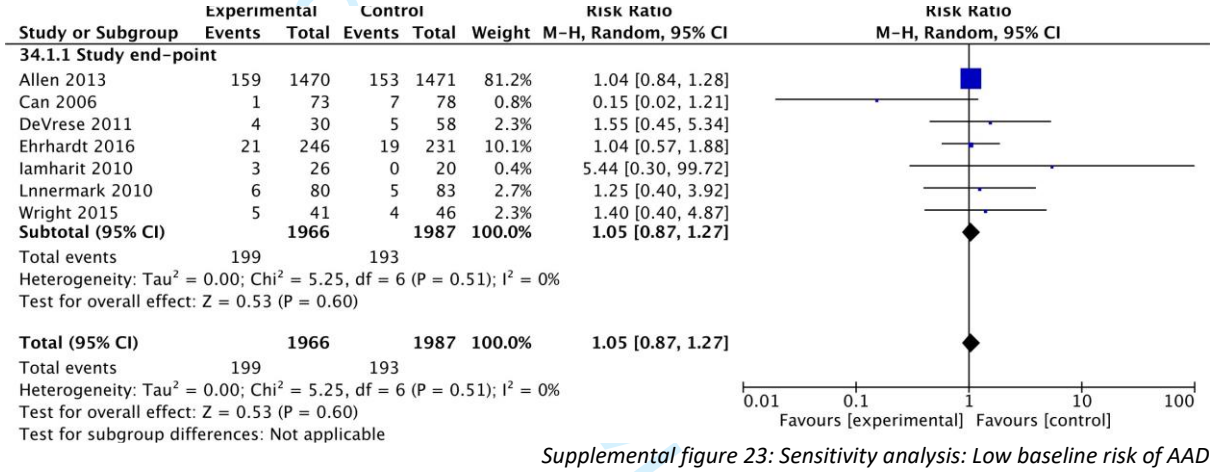
Supplemental figure 20: Subgroup analysis: species: C. butyricum vs any control



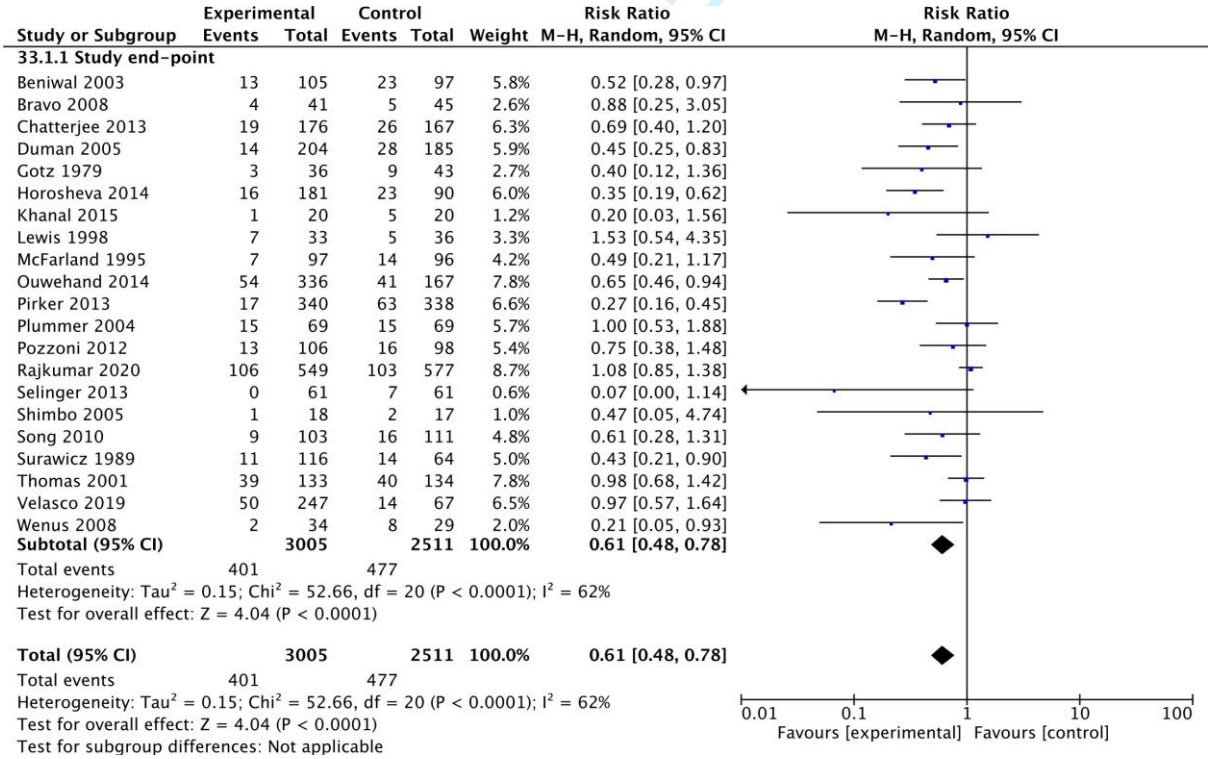
Supplemental figure 21: Subgroup analysis: species: Bacillus clausii vs any control



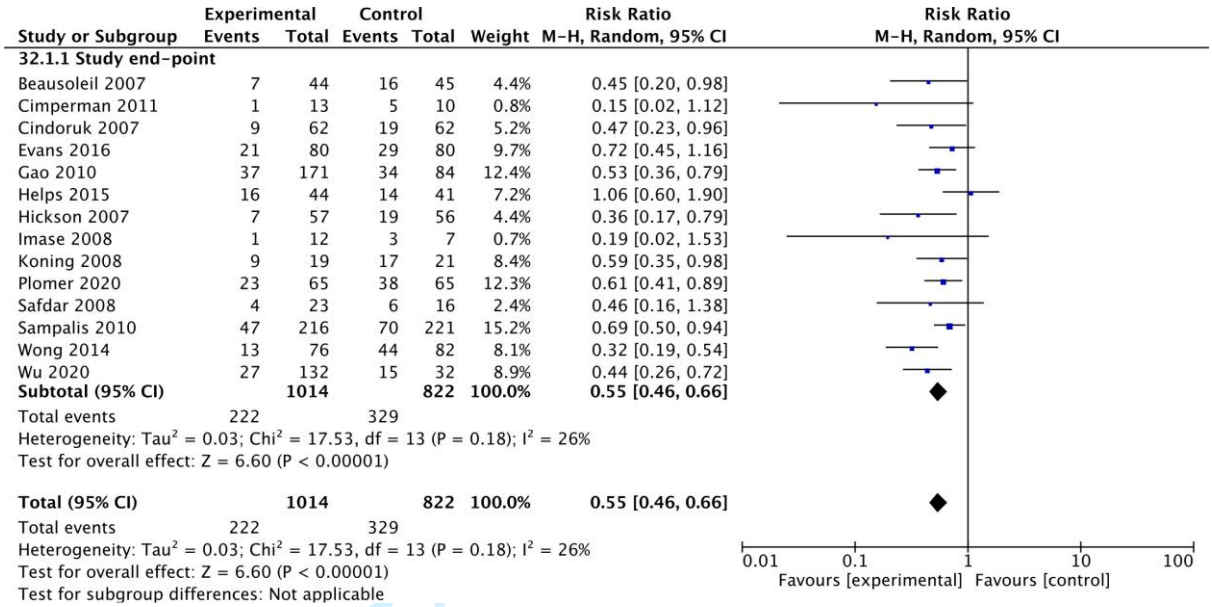
Supplemental figure 22: Subgroup analysis: Probiotics administered concurrently with H. pylori eradication therapy



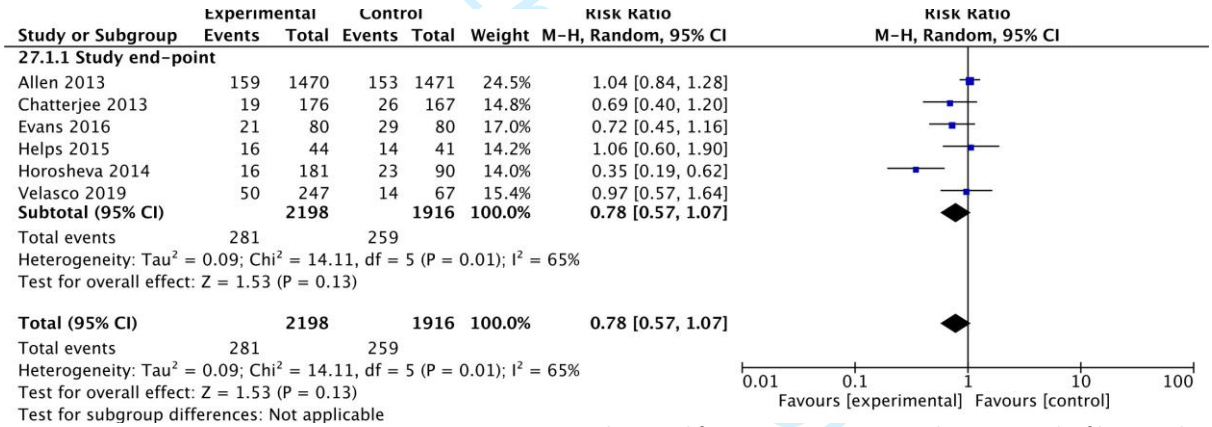
Supplemental figure 23: Sensitivity analysis: Low baseline risk of AAD



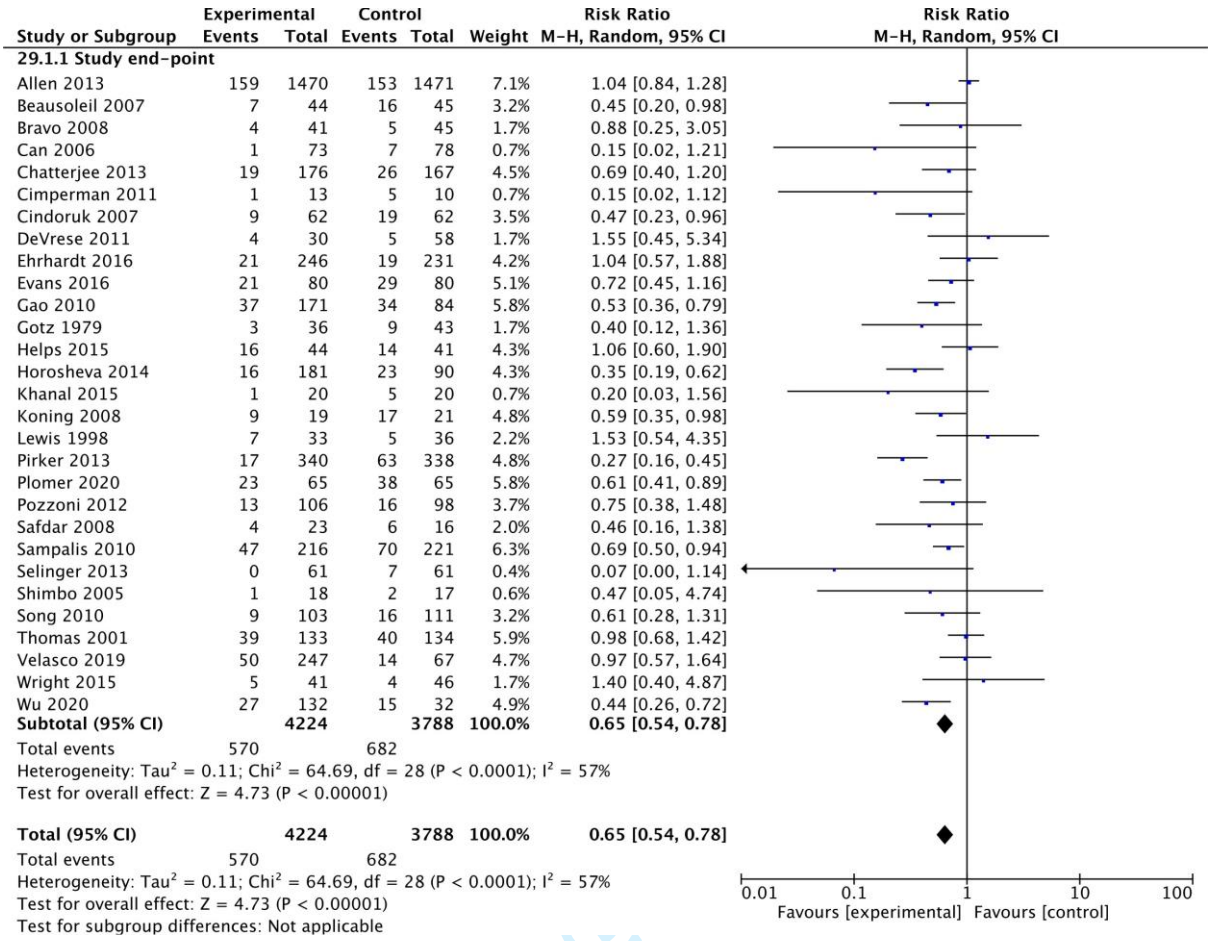
Supplemental figure 24: Sensitivity analysis: Moderate baseline risk of AAD



Supplemental figure 25: Sensitivity analysis: High baseline risk of AAD

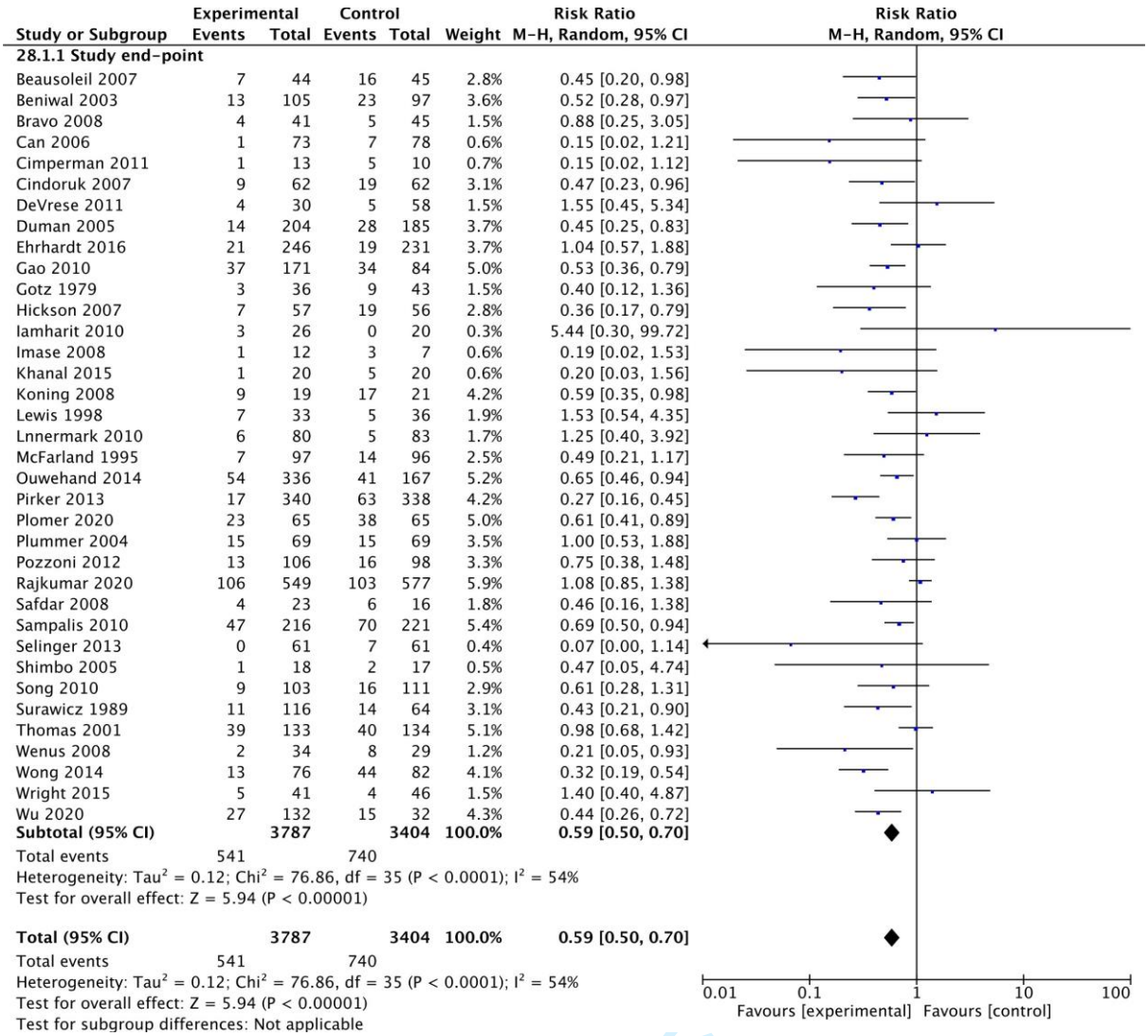


Supplemental figure 26: Sensitivity analysis: Low risk of bias studies



Supplemental figure 27: Sensitivity analysis: Low/unknown risk of bias studies

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Supplemental figure 28: High/unknown risk of bias studies

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	4
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	4
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	Appendix
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	5
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	5
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	5 / Appendix
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	5
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	5
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	5
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	5
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	5
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	5

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	5
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	5
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	6
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	6
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	8
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	8
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	8
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	9
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	9
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	10
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	11
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	12
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	13

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(6): e1000097. doi:10.1371/journal.pmed1000097

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